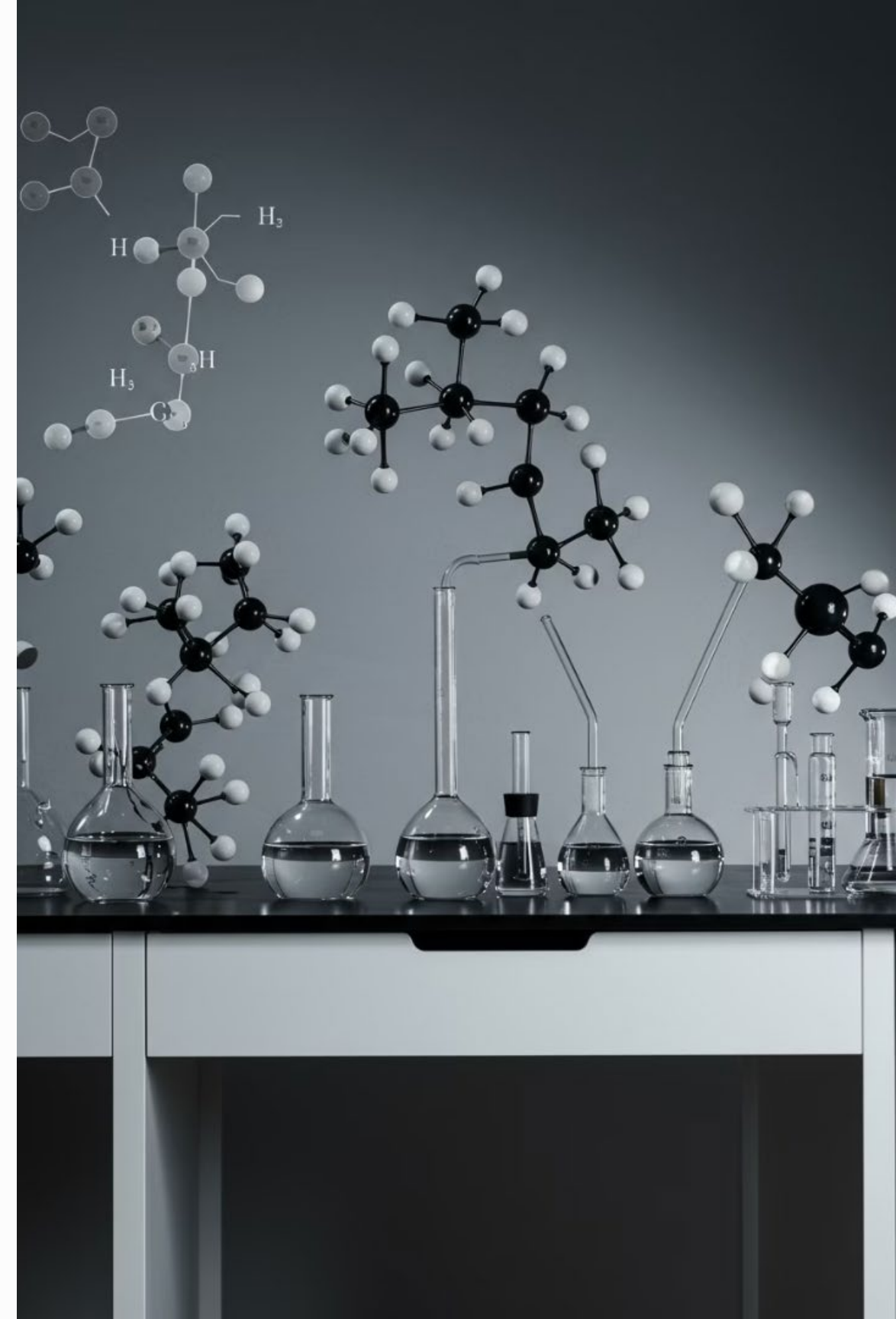


Organic Chemistry KE535

Welcome to Organic Chemistry KE535 and KE560! This course covers fundamental concepts in organic chemistry, focusing on carbon-based compounds and their reactions. Course materials include chapters 16-19 and 21-24 from Blackman's "Chemistry" textbook.

The course is led by Associate Professor Changzhu Wu, with many students supporting E-hours and laboratory sessions (KE535).

 by Changzhu Wu





Course Organization and Communication



Communication Channel

All communication will take place in English via the Itslearning platform chat function. Email responses cannot be guaranteed.



Study Materials

Essential materials include the Blackman textbook, ChemDraw software (installation instructions on Itslearning), and a molecular model kit.



Active Participation

Questions are strongly encouraged during lectures to ensure everyone understands the concepts effectively.

Course Success Strategies



Read the Textbook Thoroughly

Lecture slides supplement but don't replace textbook content



Practice Drawing Molecules

Use both hand-drawing and ChemDraw software



Participate in E-hours

Complete weekly practice questions and E-tests

For KE535, Laboratory session 3 is scheduled for **May 11**, and the trial exam is mandatory for all students.

For KE560, no laboratory sections but 2 additional lectures are given.

Why Organic Chemistry Matters in Biology and Biomedicine

Life Processes

Organic molecules form the basis of all metabolic pathways. They enable energy production and cellular communication.

Metabolism

Enzymes interact with organic compounds to drive biochemical reactions. This facilitates growth, reproduction, and homeostasis.

Pharmacology

Drug design relies on understanding organic structures. Most medications work by interacting with specific biological targets.

Genetics

DNA structure depends on organic chemistry principles. Nucleotides connect through specific bonding patterns.





Chapter 16: The Chemistry of Carbon



C-H Frameworks

Alkanes, cycloalkanes, alkenes, alkynes, and arenes



Physical Properties

Nonpolar compounds with weak dispersion forces



Isomerism

Different structural arrangements with distinct properties



Reaction Patterns

Characteristic additions and substitutions

Chapter 16 provides the foundation for understanding organic chemistry through carbon's ability to form diverse hydrocarbon frameworks.

Introduction to Hydrocarbons



Alkanes

Single bonds only (C-C)

Example: methane (CH_4)



Alkenes

Contains double bonds (C=C)

Example: ethene (C_2H_4)



Alkynes

Contains triple bonds (C $\equiv$ C)

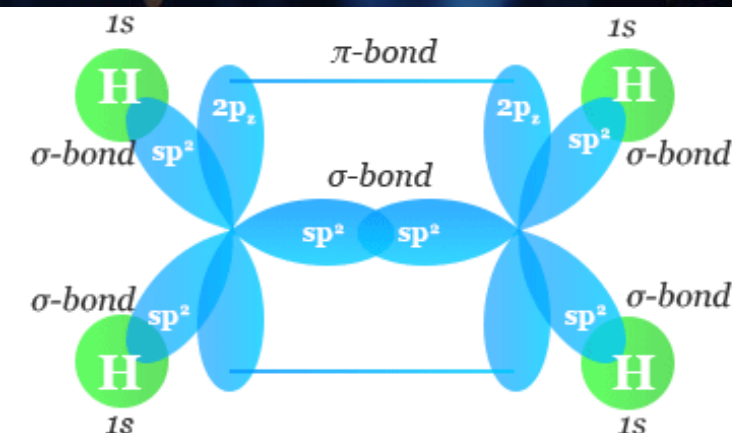
Example: ethyne (C_2H_2)



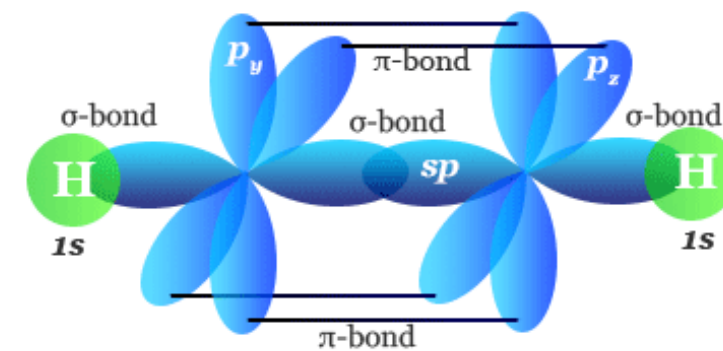
Arenes

Aromatic ring structures

Example: benzene (C_6H_6)



Structure of ethylene (C_2H_4)



Structure of acetylene (C_2H_2)

Key Concepts in Hydrocarbon Chemistry

Alkanes and Cycloalkanes

Contain only single bonds, relatively unreactive. Characterized by tetrahedral geometry and approximate 109.5° bond angles.

Alkenes and Alkynes

Contain carbon-carbon multiple bonds. Undergo addition reactions across these unsaturated bonds.

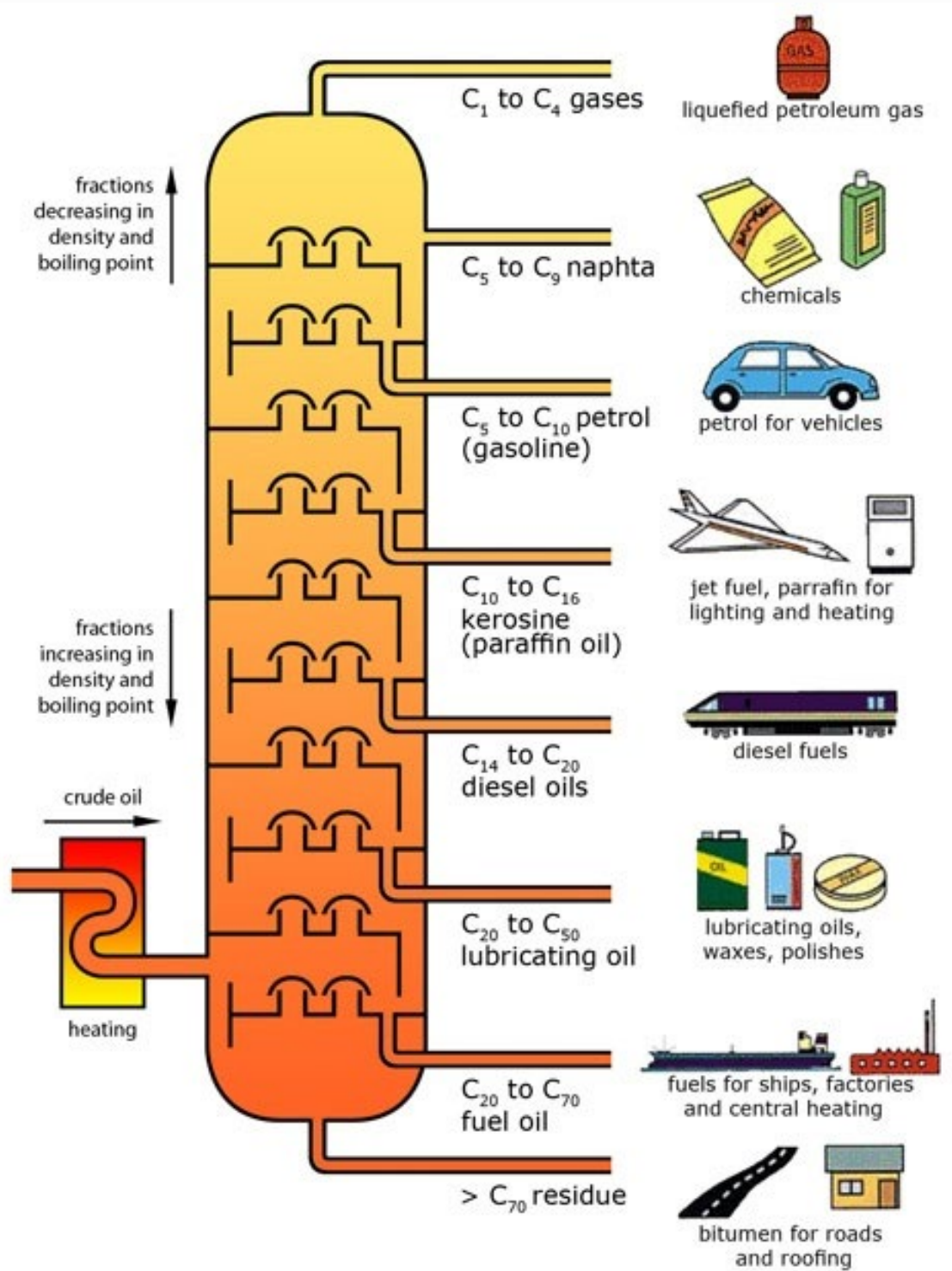
Aromatic Compounds

Feature resonance-stabilized ring structures. React through electrophilic aromatic substitution rather than addition.

Isomerism

Different structural arrangements of the same molecular formula create compounds with distinct physical and chemical properties.

Natural Sources of Hydrocarbons



Natural Sources of Hydrocarbons

7,000+

Years

Time for fossil fuel formation from ancient biomass

85%

Energy

Global energy derived from hydrocarbon sources

5,000+

Products

Different items made from petroleum derivatives

65%

Carbon

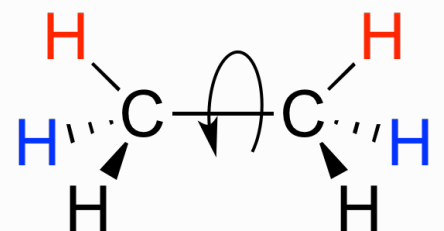
Proportion of crude oil composed of alkanes

Molecular Motion in Hydrocarbons



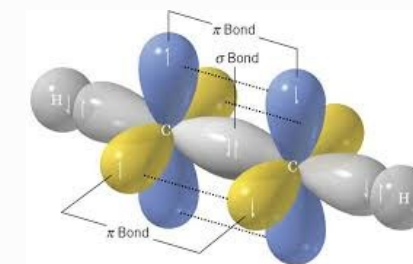
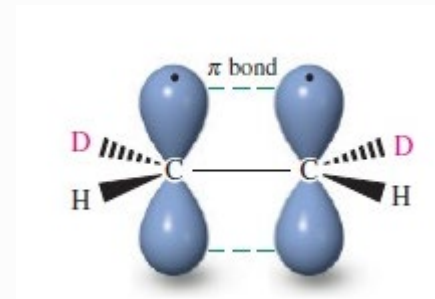
Bond Rotation

Single bonds allow free rotation, creating different conformations of the same molecule.



Restricted Rotation

Double and triple bonds prevent rotation, leading to geometric isomerism.

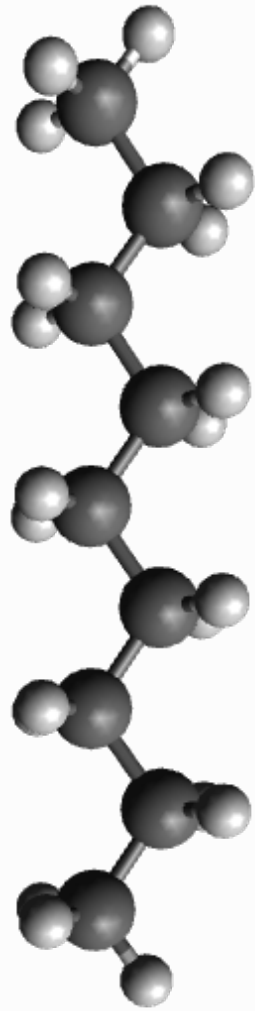


Energy Barriers

Different conformational states have varying energy levels, affecting molecular stability.

Understanding molecular motion is crucial for predicting reactions and molecular properties. The dynamic nature of hydrocarbon structures influences their behavior in chemical processes.

Alkanes (C-C): Structure and Properties



1

Tetrahedral Geometry

109.5° bond angles around carbon

2

Single C-C Bonds

Strong sigma bonds from orbital overlap

3

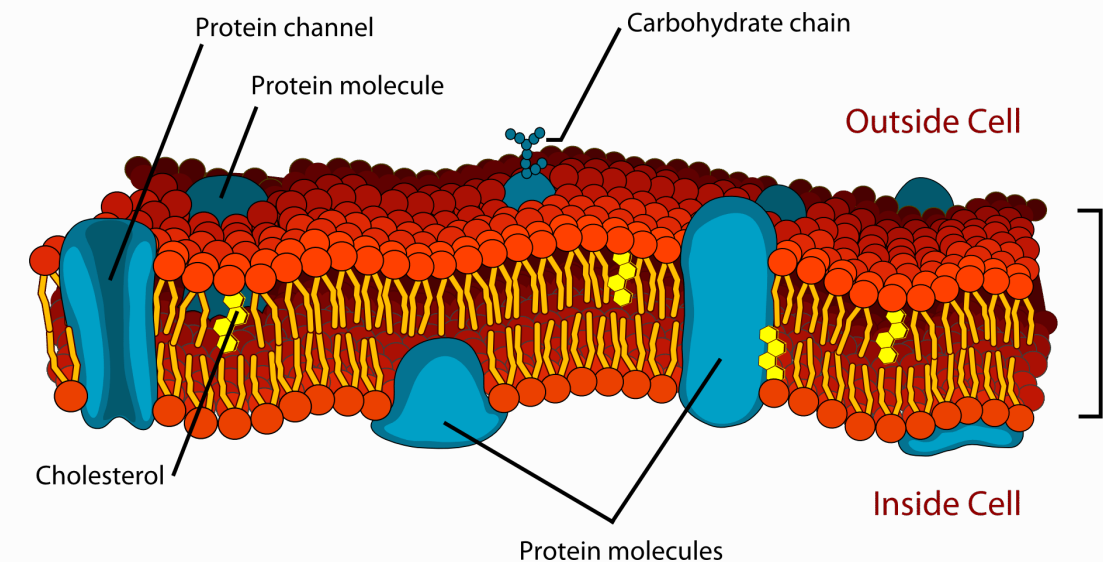
General Formula

C_nH_{2n+2} where n is carbon count

4

Biological Relevance

Form hydrophobic regions in membranes and proteins



Alkane Nomenclature and Properties

TABLE 16.1 Names, molecular formulae and condensed structural formulae for the first 10 straight-chain alkanes					
Name	Molecular formula	Condensed structural formula	Melting point (°C)	Boiling point (°C)	Density of liquid (g mL ⁻¹ at 0 °C) ^(a)
methane	CH ₄	CH ₄	−182	−164	(a gas)
ethane	C ₂ H ₆	CH ₃ CH ₃	−183	−88	(a gas)
propane	C ₃ H ₈	CH ₃ CH ₂ CH ₃	−190	−42	(a gas)
butane	C ₄ H ₁₀	CH ₃ (CH ₂) ₂ CH ₃	−138	0	(a gas)
pentane	C ₅ H ₁₂	CH ₃ (CH ₂) ₃ CH ₃	−130	36	0.626
hexane	C ₆ H ₁₄	CH ₃ (CH ₂) ₄ CH ₃	−95	69	0.659
heptane	C ₇ H ₁₆	CH ₃ (CH ₂) ₅ CH ₃	−90	98	0.684
octane	C ₈ H ₁₈	CH ₃ (CH ₂) ₆ CH ₃	−57	126	0.703
nonane	C ₉ H ₂₀	CH ₃ (CH ₂) ₇ CH ₃	−51	151	0.718
decane	C ₁₀ H ₂₂	CH ₃ (CH ₂) ₈ CH ₃	−30	174	0.730

^(a)For comparison, the density of H₂O is 1 g mL⁻¹ at 4 °C.

Alkane	Formula	Boiling Point (°C)	Physical State (25°C)
Methane	CH ₄	-161	Gas
Ethane	C ₂ H ₆	-88	Gas
Propane	C ₃ H ₈	-42	Gas
Butane	C ₄ H ₁₀	0	Gas
Pentane	C ₅ H ₁₂	36	Liquid

Alkanes are named with the suffix **"-ane"** and follow the general formula C_nH_{2n+2}. **Physical properties such as boiling point increase with molecular weight due to stronger dispersion forces.**



Alkane Structure

FIGURE 16.8

Structures of propane, butane and pentane

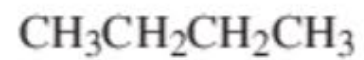
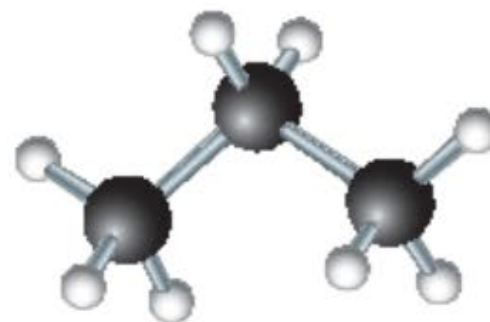
Condensed structural
formula

Line structure

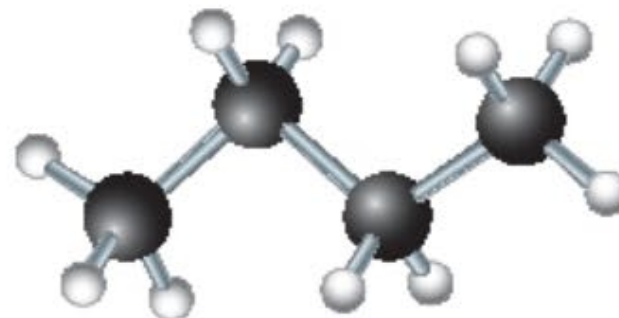
Ball-and-stick model



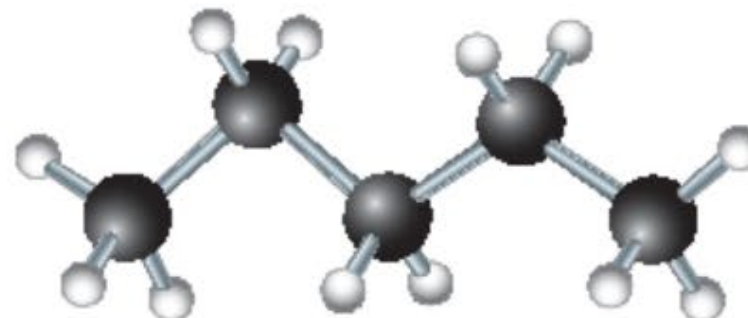
propane



butane



pentane



IUPAC Nomenclature of Alkanes

1

Identify Longest Chain

Find the longest continuous carbon chain to determine the parent name

2

Number Carbon Atoms

Start from the end closest to the first substituent

3

Name Substituents

Identify and alphabetize all side groups with position numbers

4

Combine Terms

Write the full name with hyphens and numbers

Straight-chain alkanes:

•**Methane:** CH₄

•**Ethane:** C₂H₆

•**Propane:** C₃H₈

•**Butane:** C₄H₁₀

•**Pentane:** C₅H₁₂

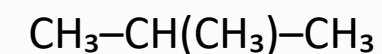
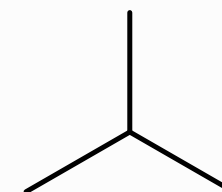
•**Hexane:** C₆H₁₄

•**Heptane:** C₇H₁₆

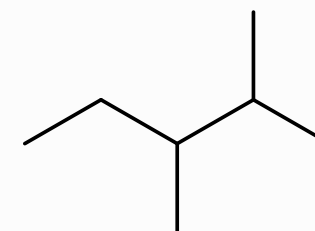
•**Octane:** C₈H₁₈

•**Nonane:** C₉H₂₀

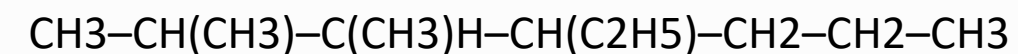
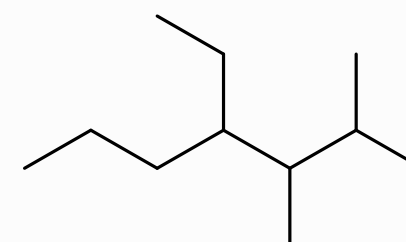
•**Decane:** C₁₀H₂₂



2-Methylpropane



2,3-Dimethylpentane



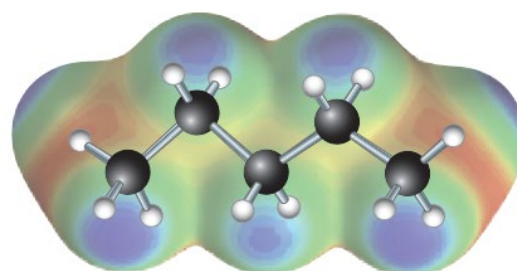
4-Ethyl-2,3-dimethylheptane

All alkane names end with the **suffix "-ane."** The first ten straight-chain alkanes from methane (CH₄) to decane (C₁₀H₂₂) form the basis for naming more complex structures.

Physical properties of alkanes

- Non-polar molecules
- Lower boiling points
- Weak dispersion intermolecular forces
- Melting point increase with molar mass
- Average density between $0.7\text{--}0.8\text{ g mL}^{-1}$
- Different properties for isomeric alkanes

FIGURE 16.21 The electron density model of pentane (like all alkanes) shows no evidence of polarity.



Alkane Conformations

Conformational Analysis

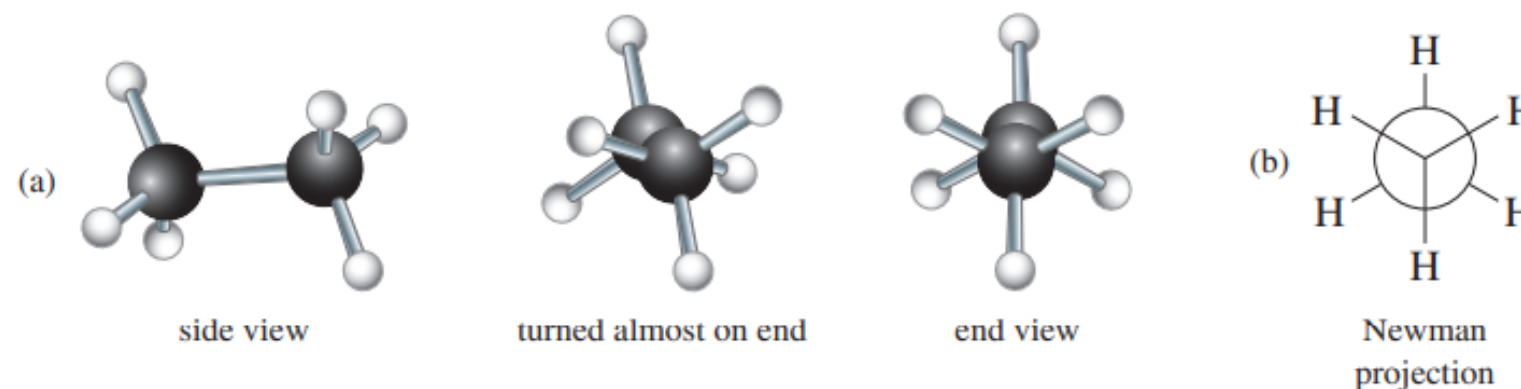
Single bonds allow rotation, creating different spatial arrangements called conformations.

These conformations differ in energy due to interactions between atoms or groups.

Newman projections provide a useful way to visualize and analyze these conformational states.

FIGURE 16.9 A staggered conformation of ethane: (a) ball-and-stick models and (b) Newman projection

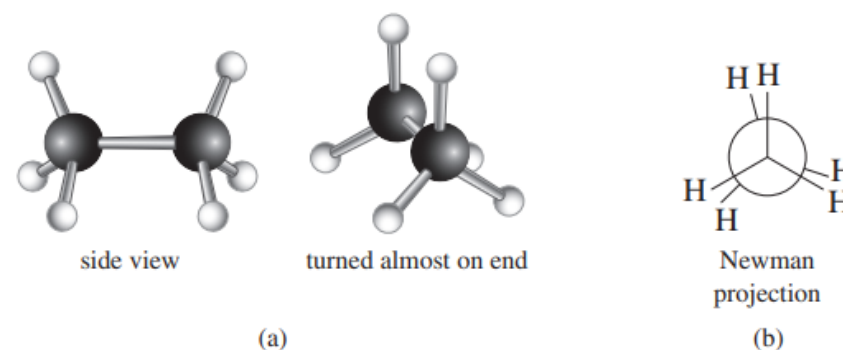
Staggered



Hydrogen atoms are maximally separated, minimizing electronic repulsion. This represents the lowest energy state.

Eclipsed

FIGURE 16.10 An eclipsed conformation of ethane: (a) ball-and-stick models and (b) Newman projection, with the bonds offset slightly for clarity



Hydrogen atoms are aligned, causing electronic repulsion. This higher energy state is less stable.

Cycloalkanes: Ring Structures



Ring Formation

Carbon atoms connected in closed loop with single bonds



General Formula

C_nH_{2n} , with two fewer hydrogens than equivalent alkanes



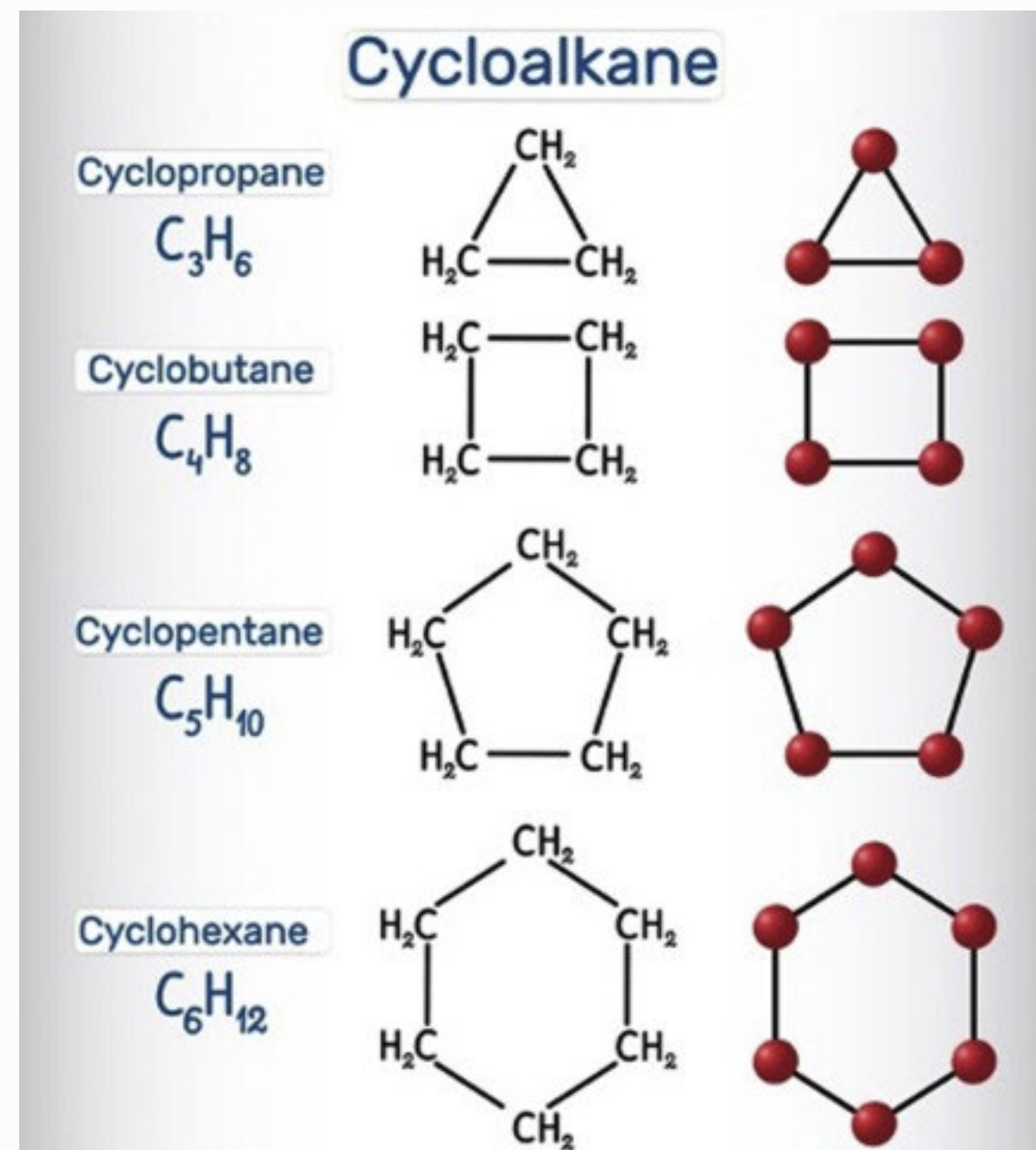
Nomenclature

Prefix "cyclo-" followed by corresponding alkane name



Stability

Five and six-membered rings most stable in biological systems



3D Structure of Cycloalkanes in Biomolecules

Chair Conformation

Most stable form of cyclohexane

Minimizes angle and torsional strain

Common in glucose and other sugars

Boat Conformation

Higher energy cyclohexane form

Hydrogen-hydrogen repulsion present

Found in certain enzyme-bound states

Envelope Conformation

Most stable form of cyclopentane

One carbon atom out of plane

Present in DNA ribose and steroid hormones

FIGURE 16.15 'Chair' conformation

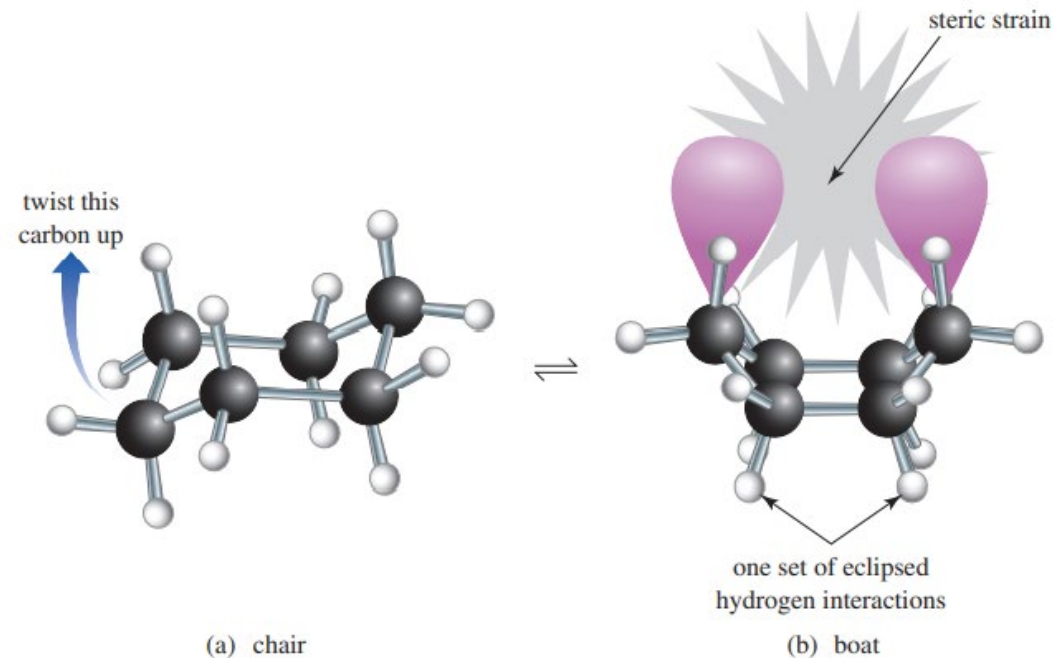
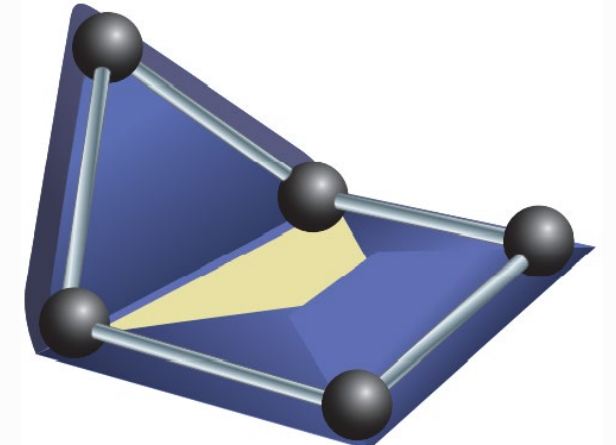
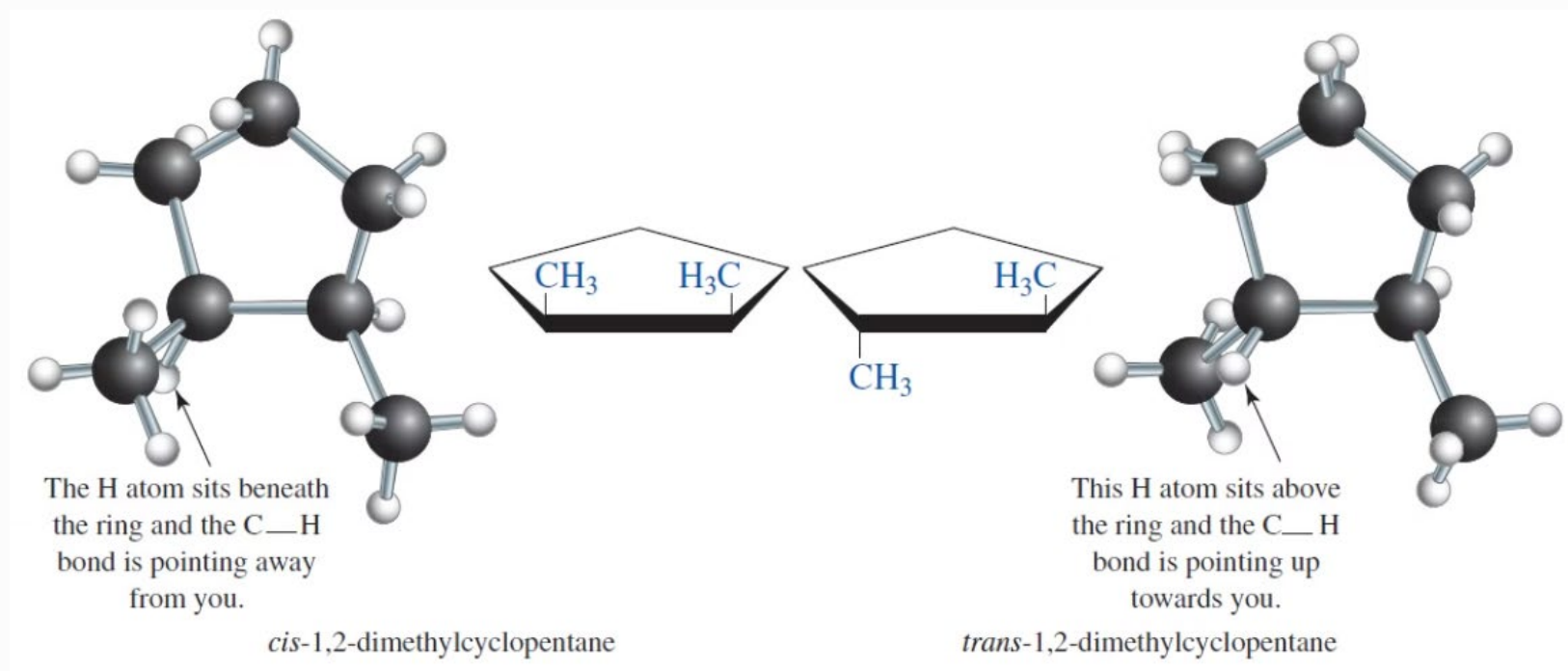


FIGURE 16.14 'Envelope' conformation



Cis-Trans Isomerism in Cycloalkanes



- **Cis Isomers**

Substituents positioned on the same side of the ring plane.

In 1,2-dimethylcyclopentane, the *cis* isomer has both methyl groups either above or below the ring.

- **Trans Isomers**

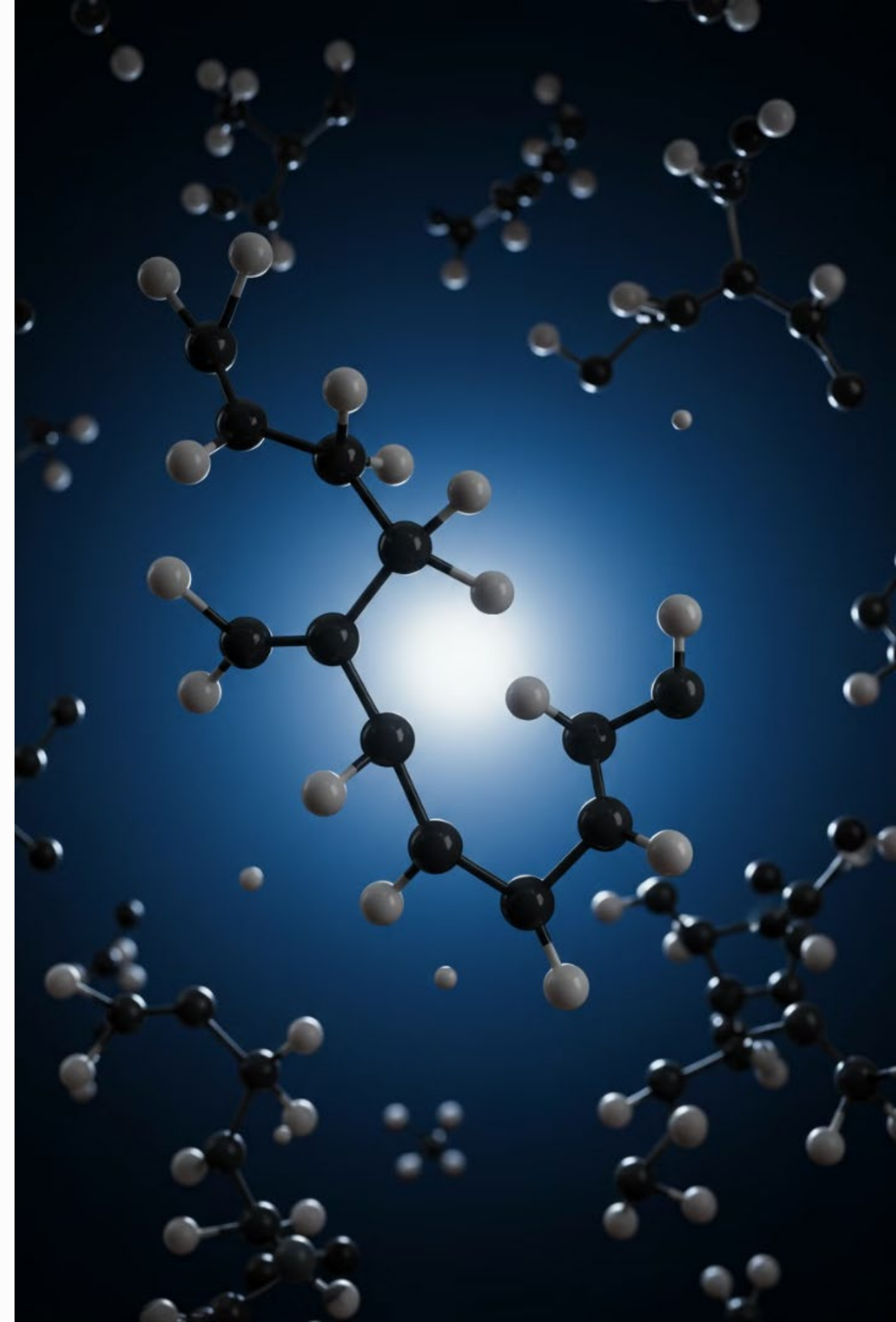
Substituents positioned on opposite sides of the ring plane.

In the *trans* isomer, one methyl group is above and one is below the ring plane.

This configurational isomerism is important because **cis and trans isomers have different physical and chemical properties**. The restricted rotation in cyclic structures prevents interconversion between these isomers under normal conditions.

Molecular Isomers: Same Formula, Different Lives

Molecular isomers are chemical compounds that share identical molecular formulas but differ in how their atoms are arranged. This seemingly small distinction creates profound differences in properties and behaviors.

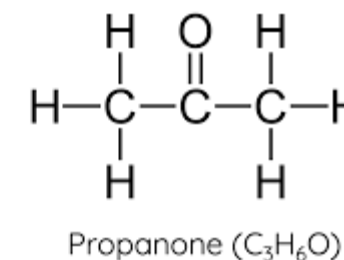
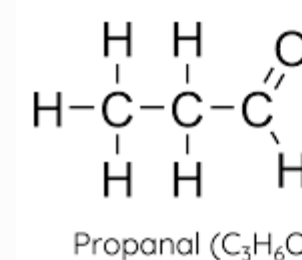
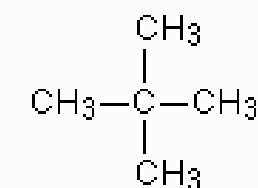
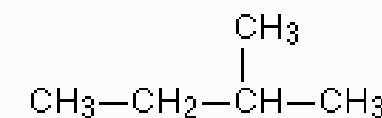
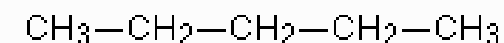


Connectivity

Example: n-butane vs. isobutane (C₄H₁₀)

Example: 1-propanol vs. 2-propanol

Example: propanal vs. acetone ($\text{C}_3\text{H}_6\text{O}$)



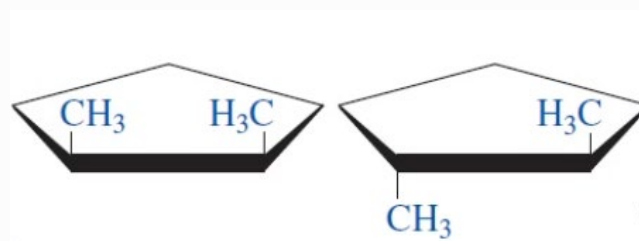
Stereoisomers: Different 3D Arrangements

- **Conformational Isomers:** Differ by rotation around single (σ) bonds.
- **Configurational Isomers:** geometric and optical isomers

Geometric Isomers (diastereomers)

Same connectivity but different spatial arrangement around **double bonds and cyclic molecules**.

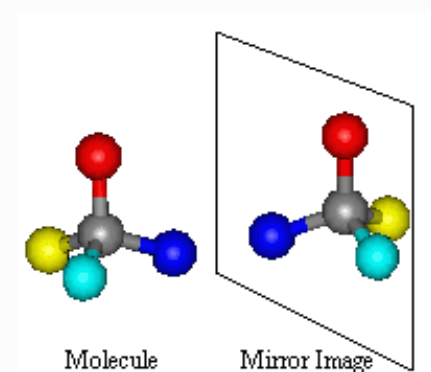
- Restricted rotation around double bonds
- Classic example: cis-trans 2-butene
- **Different physical properties**



Optical Isomers (Enantiomers)

Mirror images that cannot be superimposed.

- Contains chiral centers
- Rotate plane-polarized light differently
- Same physical properties but critical in biological systems



Real-World Impact of Isomers

Pharmaceutical Applications

One isomer may heal while another harms. Thalidomide tragedy taught us this lesson. Modern drug design targets specific isomers.

Biological Significance

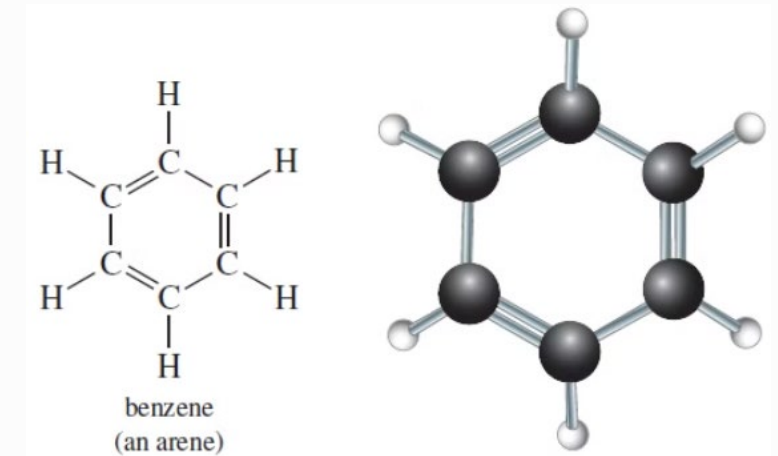
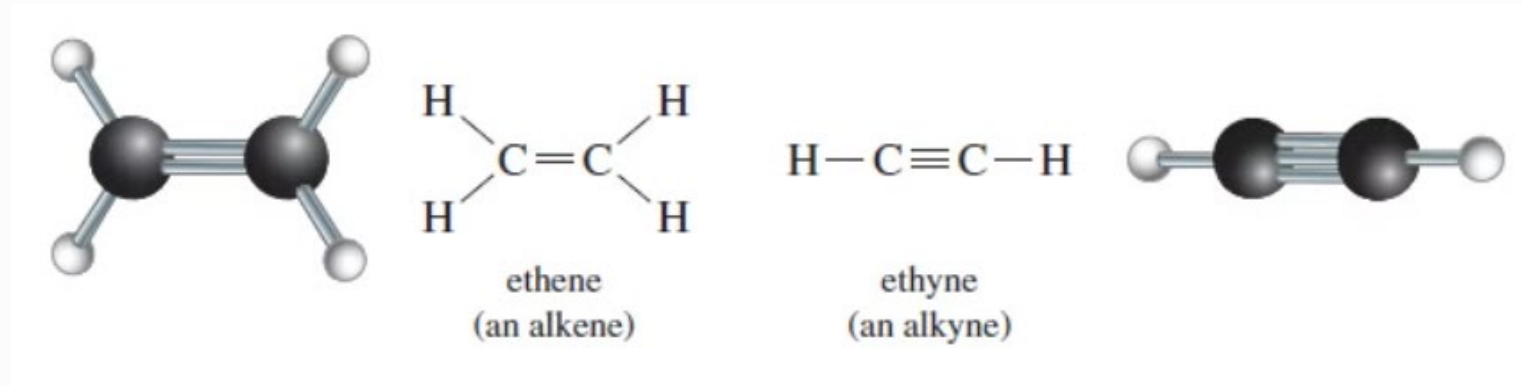
Enzymes recognize specific isomers. Our bodies metabolize L-amino acids but not their D-isomers. This selectivity drives life processes.

Industrial Importance

Petroleum refining separates valuable isomers. Fragrance industry depends on specific isomers for distinct scents. Green chemistry seeks efficient isomer synthesis.



Alkenes (C=C) and Alkynes (C≡C): Unsaturated Hydrocarbons



Alkenes (C=C)

Contain carbon-carbon double bond

General formula: C_nH_{2n}

Trigonal planar geometry (120° bond angles)

Example in biology: plant hormones like ethylene

Alkynes (C≡C)

Contain carbon-carbon triple bond

General formula: C_nH_{2n-2}

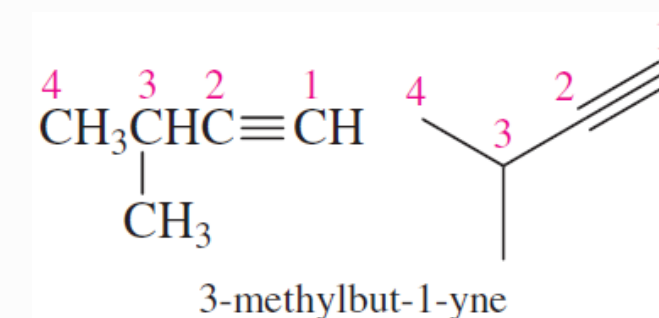
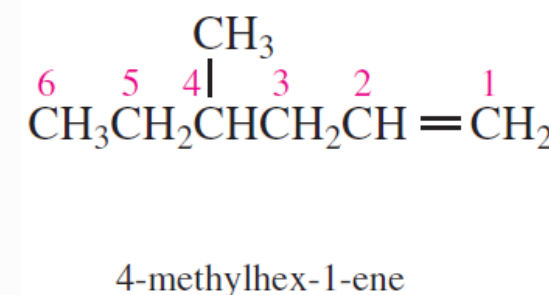
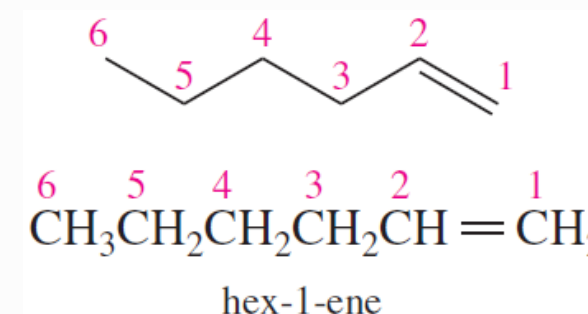
Linear geometry (180° bond angles)

Examples in biology: some antibiotics and signaling molecules

IUPAC Nomenclature of Alkenes and Alkynes

Similar rule as alkane

-  Identify Longest Chain
Must contain the multiple bond
-  Number the Chain
Start from end closer to multiple bond
-  Name Substituents
List alphabetically with position numbers
-  Specify Bond Type
Alkenes end in "-ene", alkynes end in "-yne"



Cis-Trans Isomerism in Biological Alkenes

Restricted Rotation

Double bond prevents free rotation

Creates distinct, stable geometric isomers

Cis Configuration

Similar groups on same side of double bond

Found in unsaturated fatty acids in cell membranes

Trans Configuration

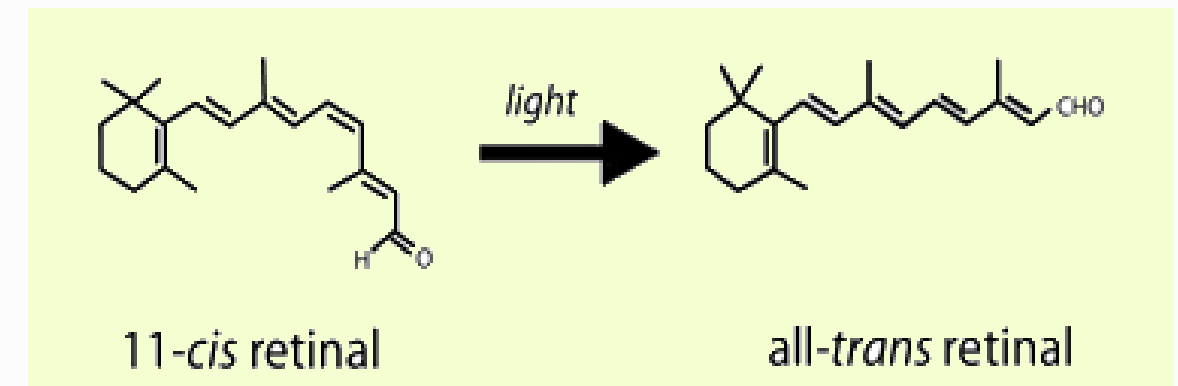
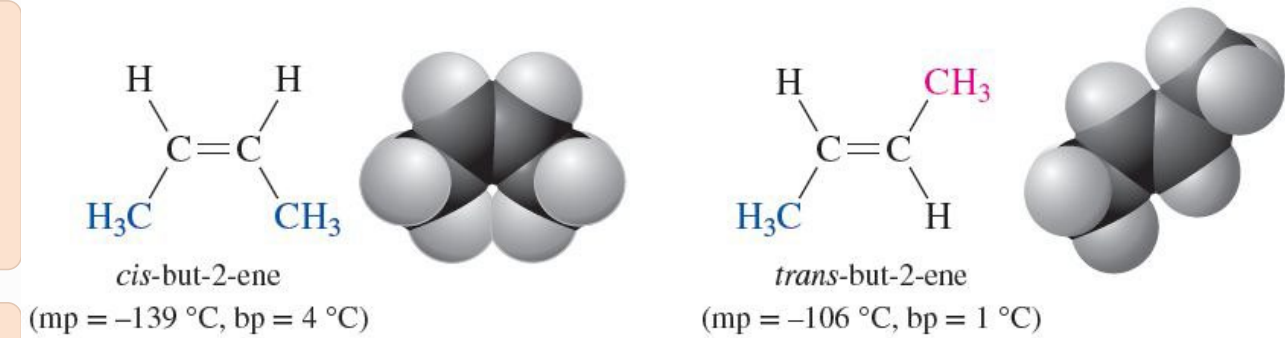
Similar groups on opposite sides of double bond

Some trans fats linked to cardiovascular disease

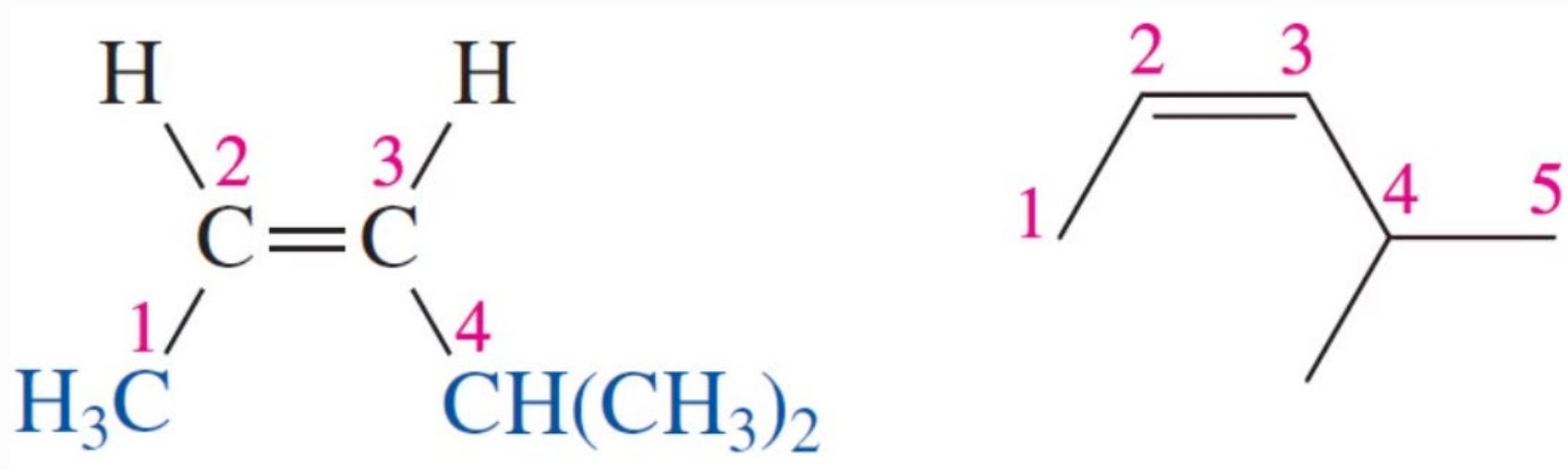
Biological Importance

Membrane fluidity depends on cis/trans ratios

Vision relies on cis-trans isomerization of retinal



E-Z Isomerism in Alkenes



Disubstituted Alkenes

When substituents on **each carbon are the same**, we use simple **cis-trans** nomenclature based on the orientation of groups in the parent chain.

Example: cis-2-butene vs. trans-2-butene.

Tri- or Tetrasubstituted Alkenes

When **substituents differ**, we use the more precise **E-Z system** based on priority rules.

- **E (entgegen):** higher priority groups on opposite sides
- **Z (zusammen):** higher priority groups on same side

Priority is determined using the Cahn-Ingold-Prelog (CIP) rules, where atoms with higher atomic numbers receive higher priority.

E-Z System for Alkene Configuration



Assign Priorities

Use **Cahn-Ingold-Prelog** rules based on atomic number



Compare Positions

Determine placement of high-priority groups



Assign Configuration

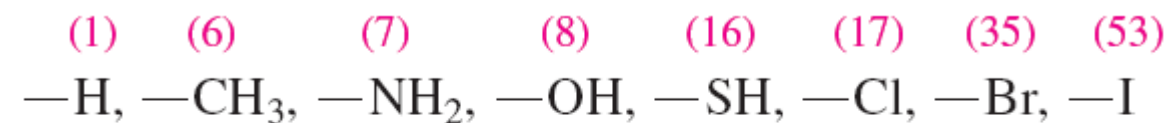
Z: higher priority groups on same side



Alternate Configuration

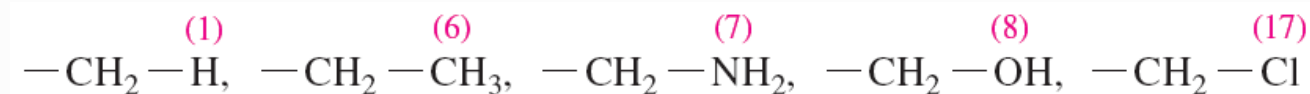
E: higher priority groups on opposite sides

- E and Z configurations priority is based on **atomic number**



Increasing priority

- Or at the first point of difference



Increasing priority

The E-Z system provides a precise method for describing the configuration of alkenes, especially when the substituents are complex or when simple cis-trans nomenclature is insufficient.

Specialized Alkene and Diene Systems

Cycloalkenes

Cyclic structures containing one or more double bonds. Small rings (≤ 7 carbons) can only exist in the cis configuration due to ring strain.

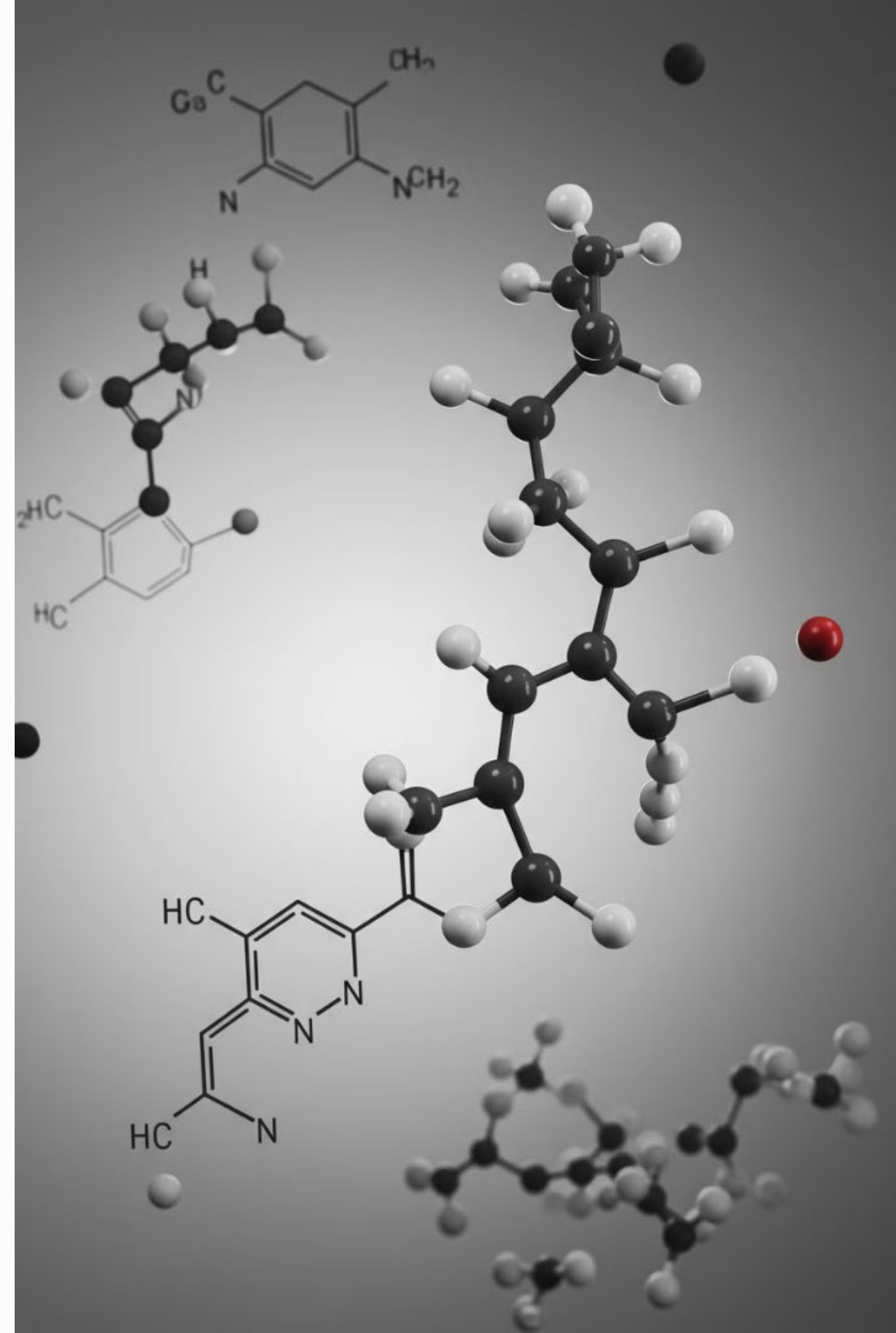
Dienes and Polyenes

Compounds with two or more double bonds. Conjugated dienes have alternating double and single bonds, providing special reactivity.

Physical Properties

Like alkanes, alkenes and dienes are nonpolar with dispersion forces as primary intermolecular interactions. They have lower density than water and are insoluble in it.

The number of possible cis-trans isomers in polyenes increases exponentially as 2^n , where n is the number of double bonds. This creates vast structural diversity in naturally occurring polyenes.



Reactions of Alkanes



Limited Reactivity

Alkanes exhibit minimal reactivity due to strong, nonpolar C-C and C-H sigma bonds.



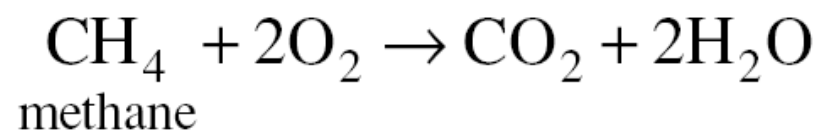
Combustion

The primary reaction of alkanes is oxidation with oxygen, producing carbon dioxide and water.

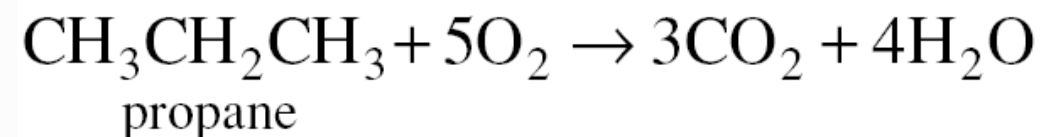


Energy Source

This highly exothermic reaction makes alkanes valuable fuels for heat and power generation.



$$\Delta H^\ominus = -886 \text{ kJ mol}^{-1}$$



$$\Delta H^\ominus = -2220 \text{ kJ mol}^{-1}$$

The general combustion reaction for alkanes can be represented as:



Reactions of Alkenes

Electrophilic Addition

The characteristic reaction of alkenes involves addition across the carbon-carbon double bond.

The π bond acts as a nucleophile, attacking electrophiles to form new sigma bonds.

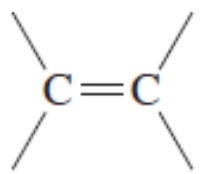
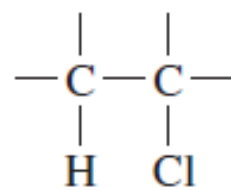
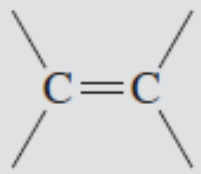
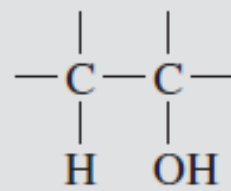
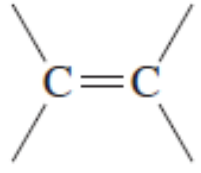
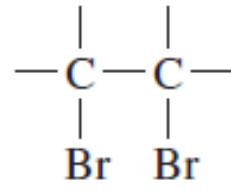
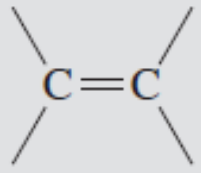
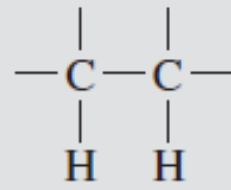
- Hydrohalogenation (HX addition)
- Halogenation (X_2 addition)
- Hydration (H_2O addition)
- Hydrogenation (H_2 addition)

Polymerization

From an industrial perspective, alkenes serve as monomers for polymer synthesis.

This process creates long chains of repeating units, resulting in valuable materials like polyethylene, polypropylene, and polyvinyl chloride.

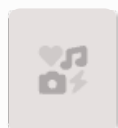
TABLE 16.4 Characteristic addition reactions of alkenes

Reaction	Descriptive name(s)
 + HCl $\longrightarrow$ 	hydrochlorination (an example of hydrohalogenation)
 + H ₂ O $\longrightarrow$ 	hydration
 + Br ₂ $\longrightarrow$ 	bromination (an example of halogenation)
 + H ₂ $\longrightarrow$ 	hydrogenation (reduction)

Reaction mechanism

- Electron Movement in Organic Reactions

Electrons: the invisible force driving chemical reactions in living systems



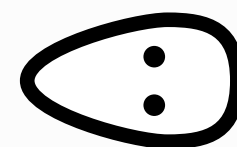
Arrow Notation

Double-barbed arrows show movement of electron pairs



Electrophiles

Electron-poor species accept electron pairs

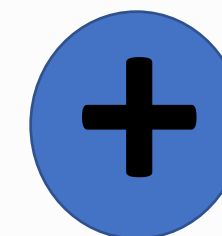


nucleophile

For making and breaking bonds



Double-barbed arrow represents the movement of two electrons



electrophile



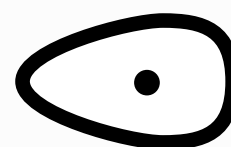
Nucleophiles

Electron-rich species attack electron-poor sites



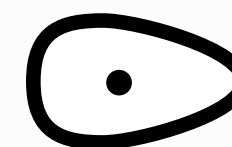
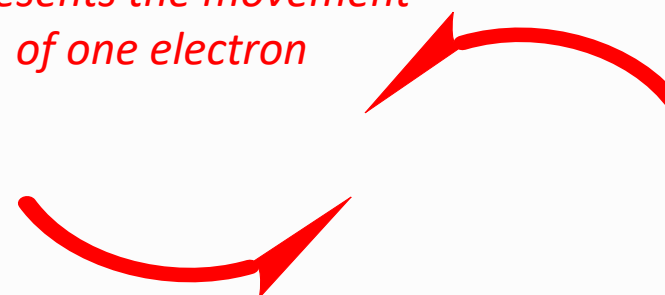
Radical Processes

Single-electron transfers in oxidative reactions



radical

Single-barbed arrow represents the movement of one electron

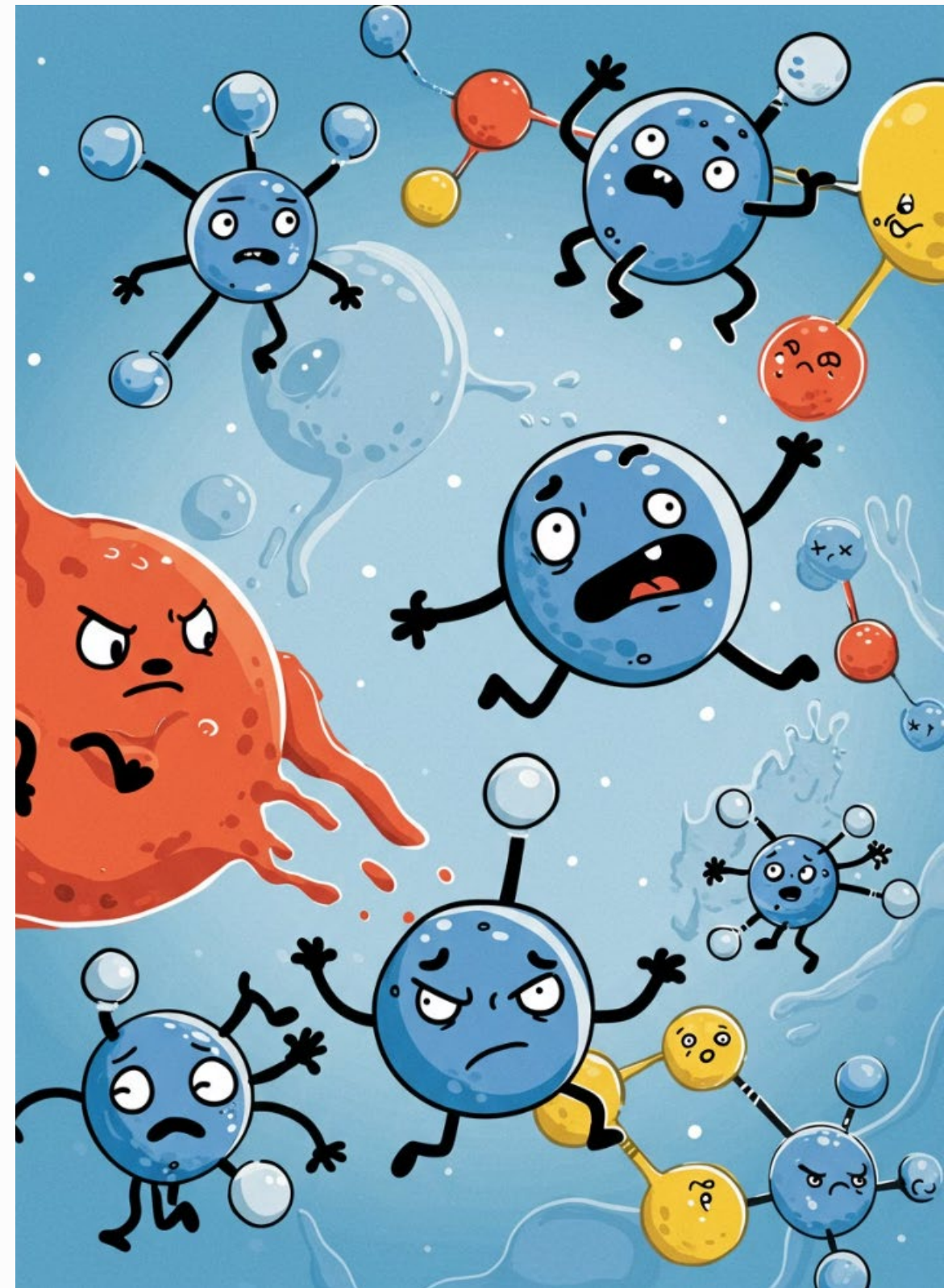


radical

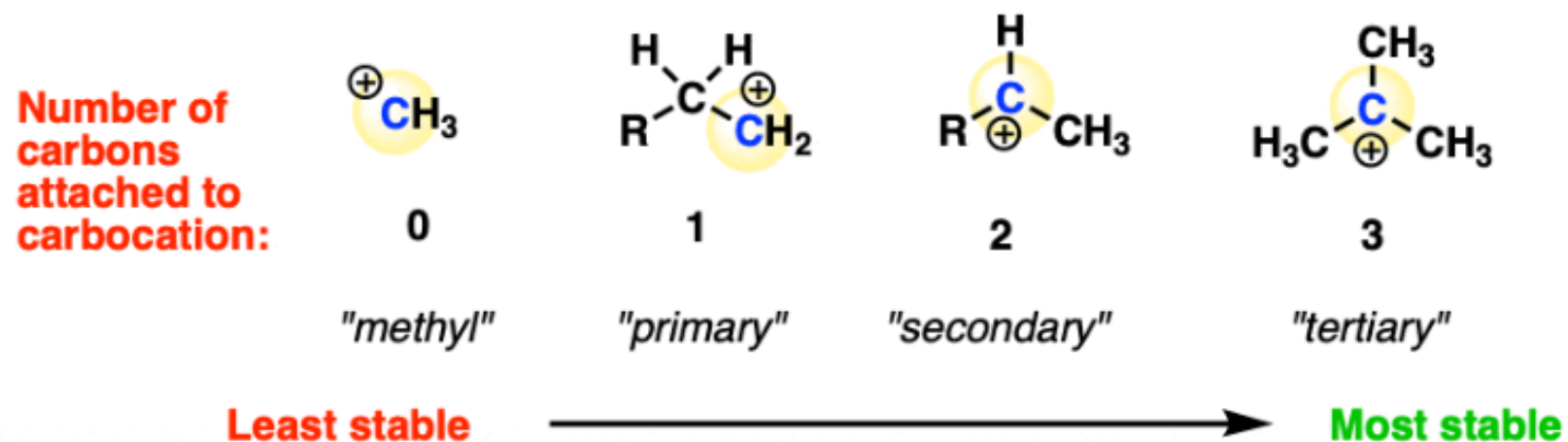
Reaction mechanism

- The Carbocation Stability

- Carbocations are carbon atoms with a positive charge and only six valence electrons.
- They're unstable and reactive because they desperately want to complete their octet!
- Stability equals survival time in reactions. More stable carbocations last longer before reacting.



Carbocation Stability Hierarchy



Tertiary (3°)

Most stable, three alkyl groups donate electrons



Secondary (2°)

Moderately stable, two alkyl groups support charge



Primary (1°)

Less stable, only one alkyl group donates electrons



Methyl (CH₃⁺)

Least stable, no alkyl groups to stabilize charge

Hydrohalogenation of Alkenes

1

Step 1: Protonation

The π electrons attack the proton from HX, forming a carbocation intermediate

2

Step 2: Halide Attack

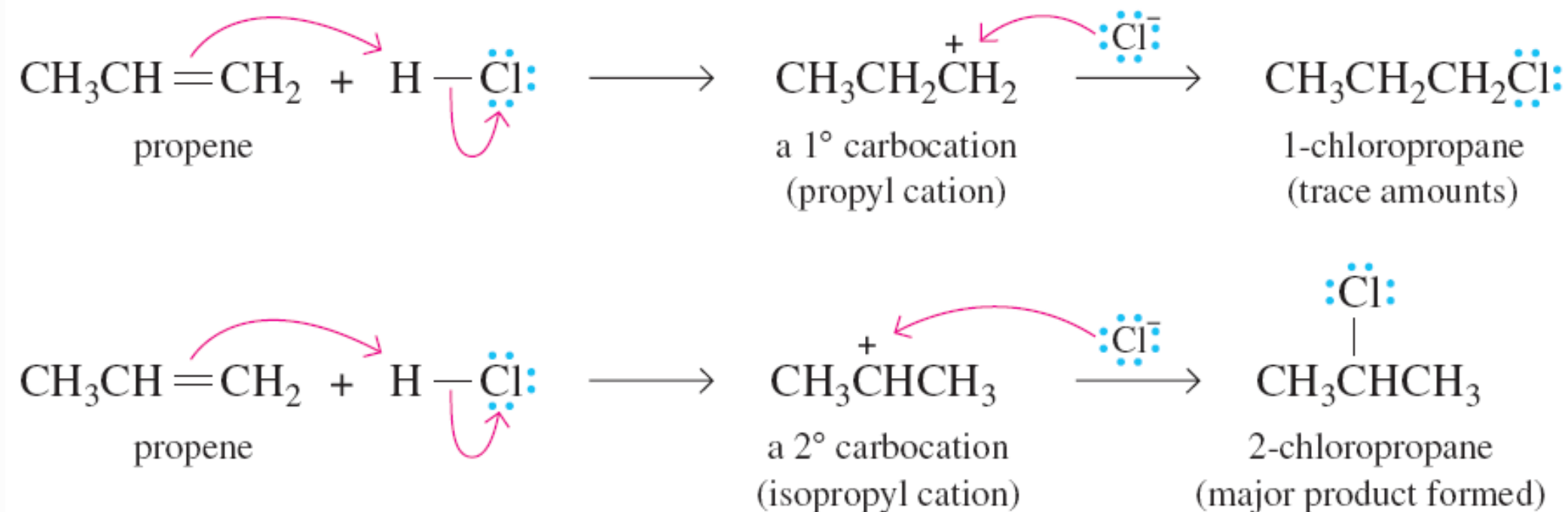
The negatively charged halide ion attacks the carbocation

3

Markovnikov's Rule

The halogen adds to the more substituted carbon atom

Carbocation stability follows the order: tertiary > secondary > primary. This explains why HX addition follows Markovnikov's rule, with H^+ adding to the less substituted carbon to form the more stable carbocation.



Halogenation of Alkenes



Halogen Polarization

The π electrons induce polarization in the approaching X_2 molecule



Cyclic Halonium Ion Formation

A three-membered cyclic halonium ion intermediate forms



Halide Attack

The halide ion attacks from the opposite side (anti addition)



Dihalide Product

The reaction produces a vicinal dihalide with anti stereochemistry

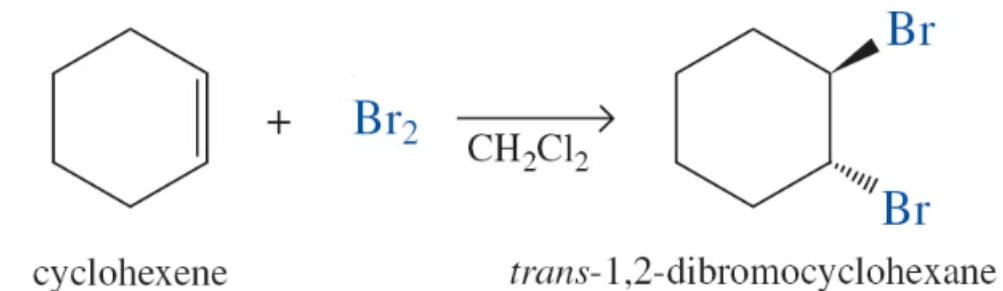
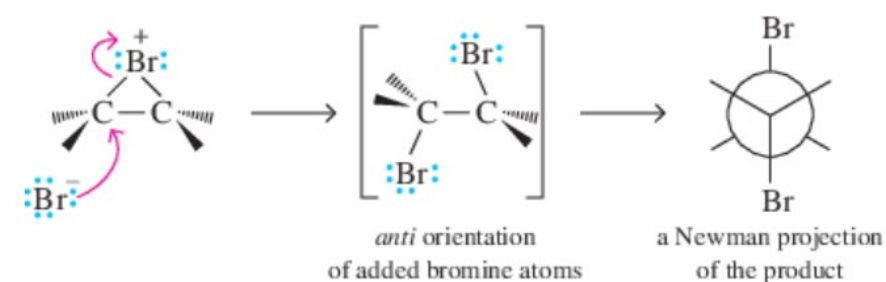
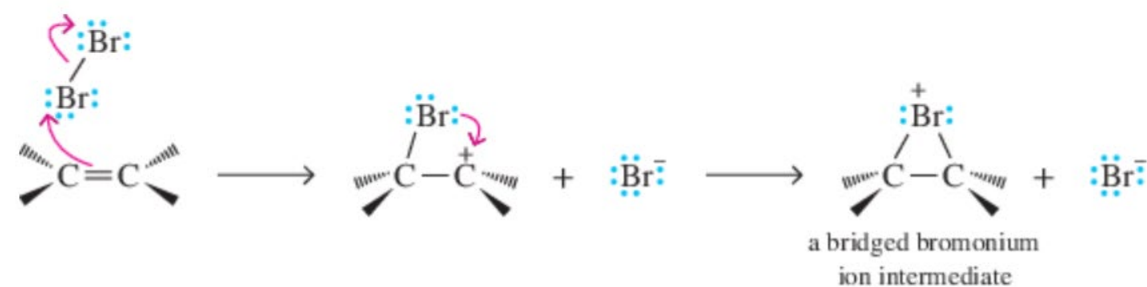
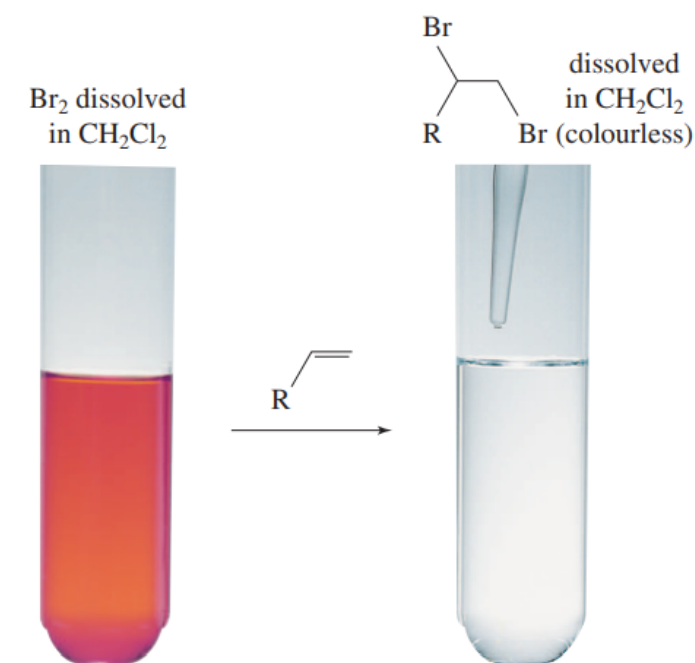


FIGURE 16.32

A solution of bromine in dichloromethane, CH_2Cl_2 , is red (left). Add a few drops of an alkene and the red colour disappears (right).



Catalytic Hydrogenation of Alkenes



Syn Addition

Both hydrogen atoms add to the same face of the alkene plane.



Thermodynamics

Hydrogenation is highly exothermic, with $\Delta H \approx -120$ kJ/mol.



Catalyst Role

Metals like Pt, Pd, or Ni adsorb H_2 , weakening the H-H bond.

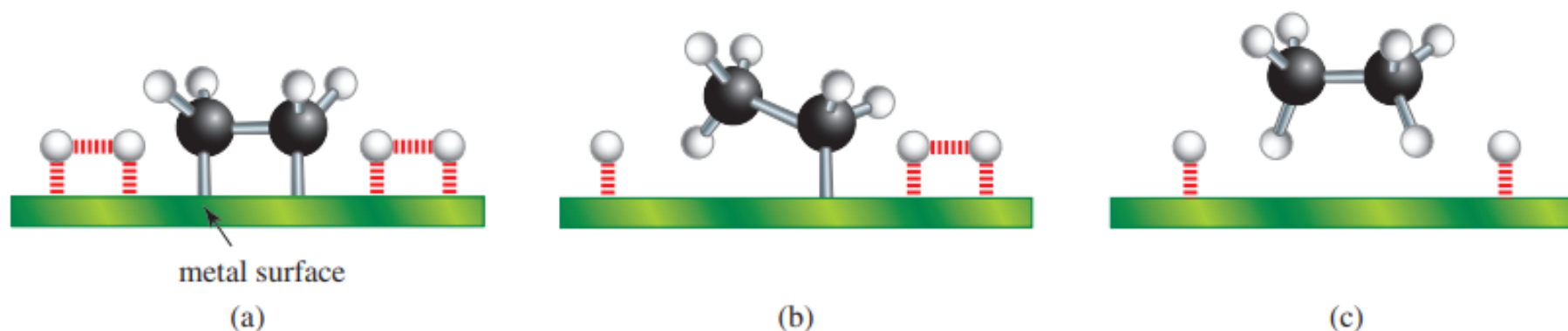


Reaction Conditions

Typically performed at room temperature and atmospheric pressure with a heterogeneous catalyst.

FIGURE 16.33

syn addition of hydrogen to an alkene involving a transition metal catalyst. (a) Hydrogen and the alkene are adsorbed on the metal surface. (b) One hydrogen atom is transferred to the alkene, forming a new C—H bond. The other carbon atom remains adsorbed on the metal surface. (c) A second C—H bond forms and the alkane is desorbed.



Catalytic hydrogenation is a fundamental reaction in organic synthesis, used for reducing double bonds in laboratory and industrial processes.

Reactions of Alkynes

Addition Reactions

Similar to alkenes, alkynes undergo electrophilic addition reactions

+

☆

Partial Reduction

Lindlar catalyst ($\text{Pd}/\text{CaCO}_3/\text{Pb}$) produces *cis*-alkenes selectively

Selective Chemistry

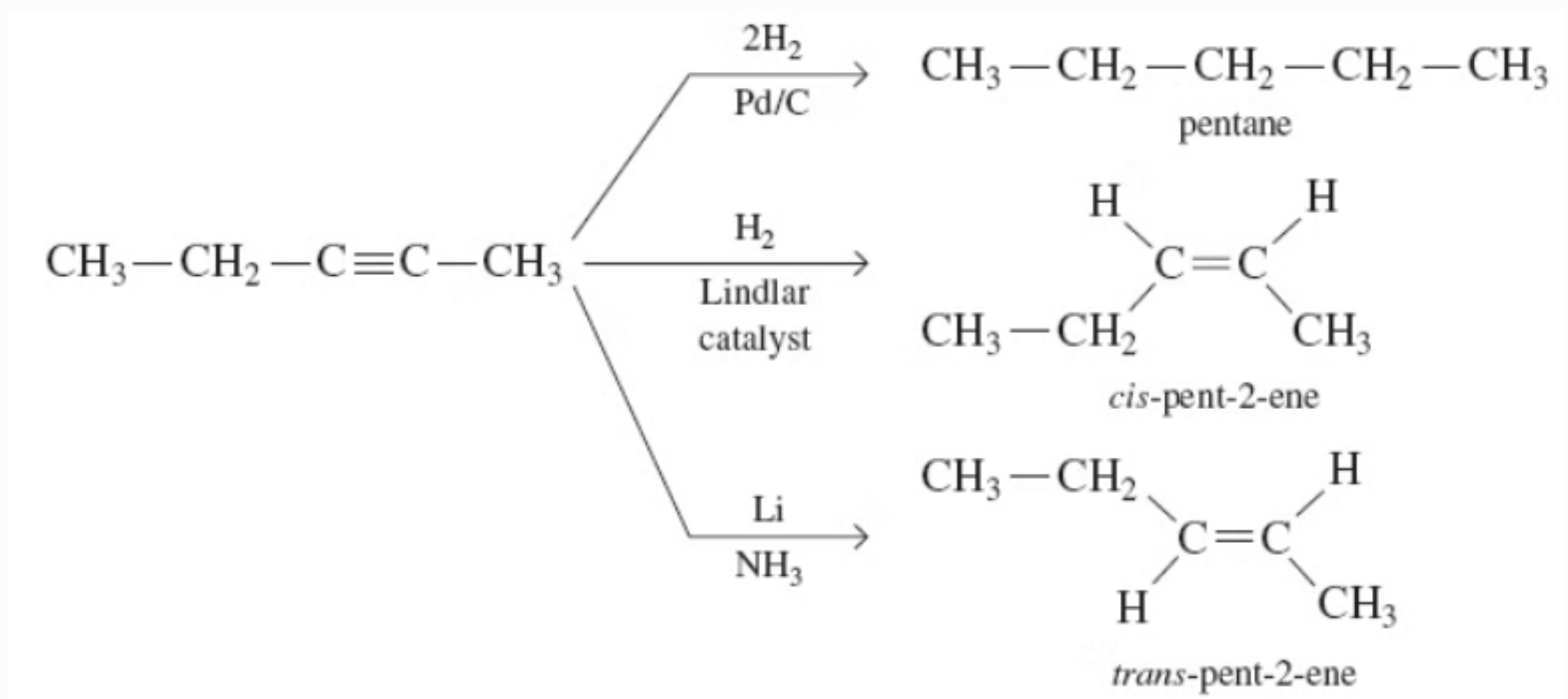
Li/NH_3 reduction produces *trans*-alkenes

📺

👤

Complete Reduction

Standard hydrogenation catalysts (Pt , Pd , Ni) produce alkanes



Benzene: Introduction to Aromatic Compounds

□ Structure and Stability

Benzene (C_6H_6) is the parent aromatic compound with a planar, hexagonal structure.

Despite being highly unsaturated, **benzene is surprisingly stable and doesn't undergo typical alkene reactions.**

□ Kekulé Structure

Friedrich August Kekulé proposed alternating single and double bonds in a hexagonal ring.

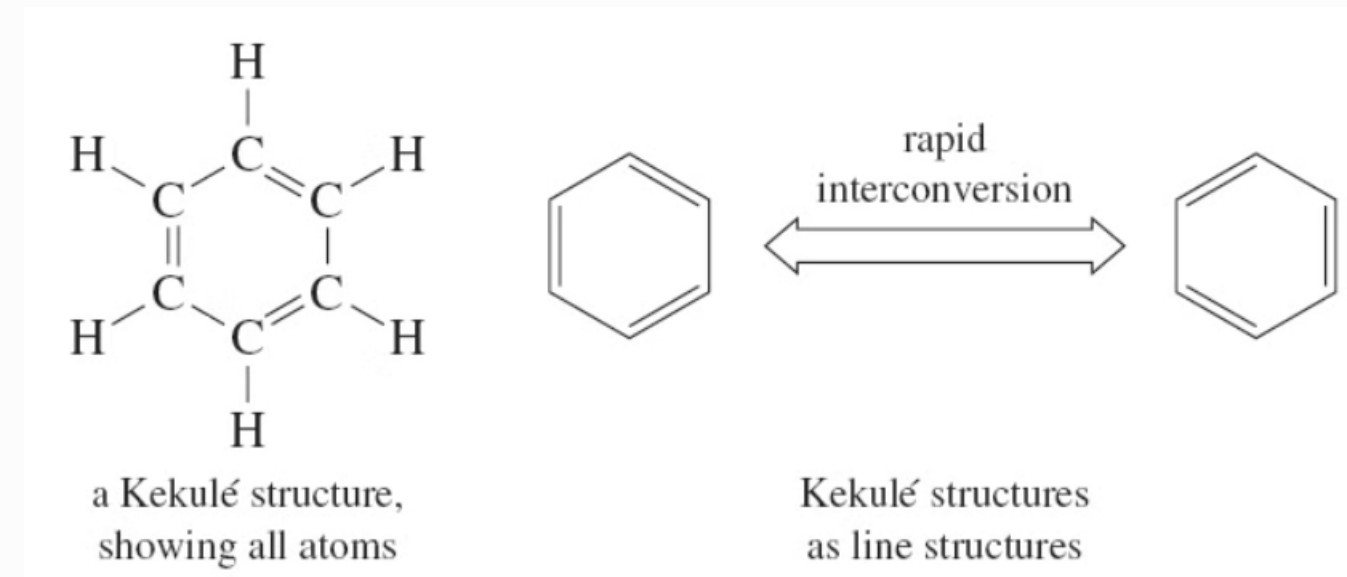
However, this model doesn't explain benzene's unusual stability and lack of alkene-like reactivity.

□ Biological Significance

Many amino acids contain aromatic rings

Aromatic compounds in DNA and RNA bases

π -stacking stabilizes protein structures



Modern understanding shows that benzene is best represented as a resonance hybrid where electrons are delocalized around the ring, creating special stability (resonance energy ≈ 150 kJ/mol).

Benzene: Resonance and Orbital Model

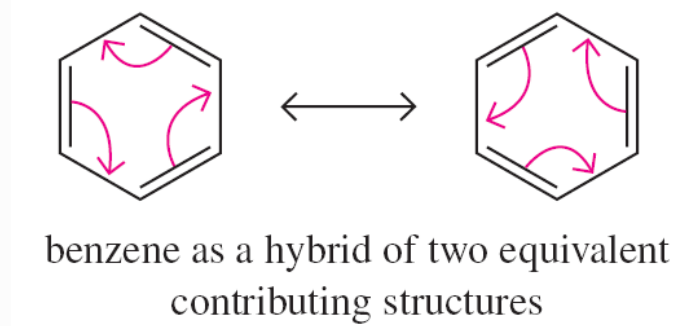
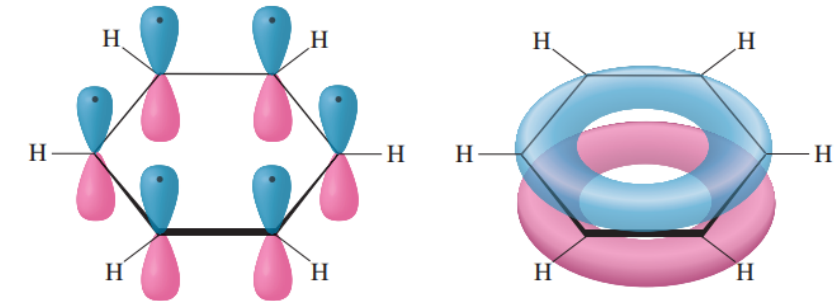


FIGURE 16.35

Orbital overlap model of bonding in benzene. (a) The carbon–hydrogen framework. The six $2p$ orbitals, each with one electron, are shown uncombined. (b) The overlap of parallel $2p$ orbitals forms a continuous π cloud, shown by one torus above the plane of the ring and a second below the plane of the ring.

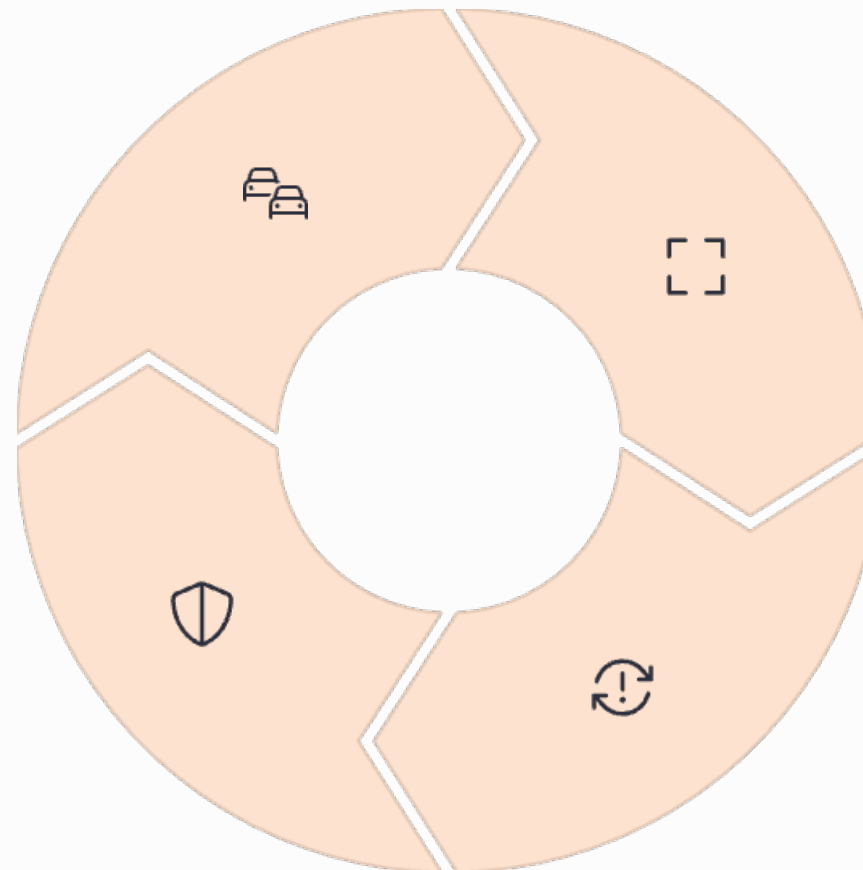


sp^2 Hybridization

Each carbon forms three sigma bonds in the plane

Aromatic Stabilization

Creates exceptional stability and distinctive reactivity



p Orbital Overlap

Forms continuous π electron cloud above and below ring

Resonance Delocalization

Six π electrons distributed across all six carbons

Hückel's Rules for Aromaticity

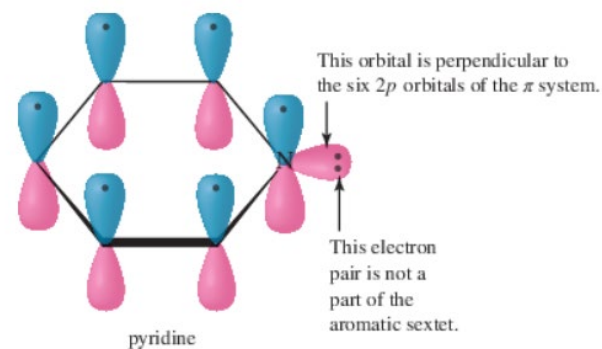
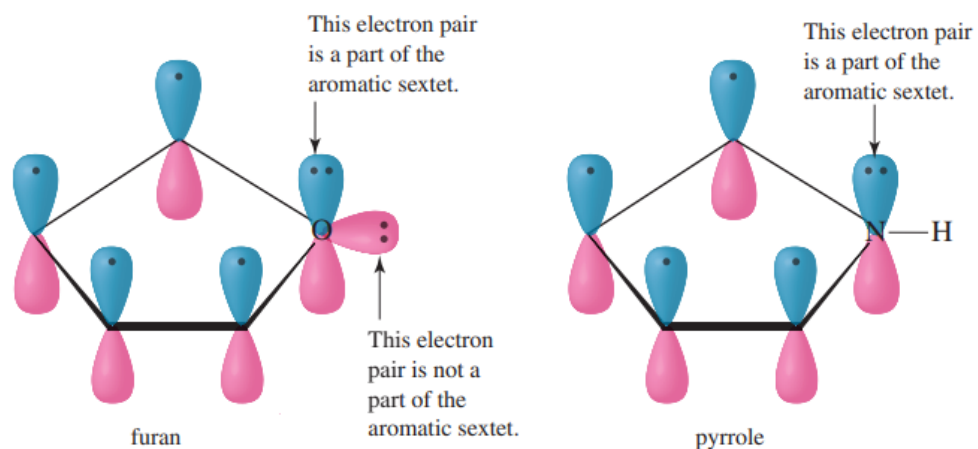


FIGURE 16.37

Origin of the six π electrons (the aromatic sextet) in furan and pyrrole. The resonance energy of furan is 67 kJ mol^{-1} and that of pyrrole is 88 kJ mol^{-1} .



1 Continuous p Orbitals

Each atom must contribute one p orbital perpendicular to ring

2 Planar Structure

Ring must be flat for effective orbital overlap

3 Electron Count

$(4n + 2)$ π electrons: 2, 6, 10, 14...

4 Complete Conjugation

All atoms must participate in the π system

Hückel's $(4n + 2)$ rule explains why benzene (6 π electrons, $n=1$), naphthalene (10 π electrons, $n=2$), and other aromatic compounds exhibit unusual stability and characteristic reactivity.

Substituted Benzene Nomenclature

Monosubstituted Benzenes

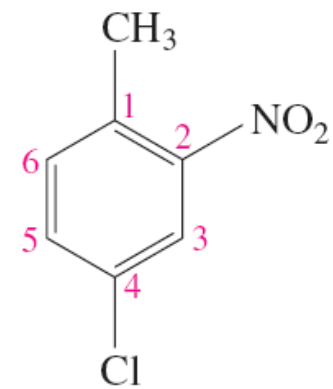
Named as derivatives of benzene or by common names (e.g., toluene, phenol, aniline).

Disubstituted Benzenes

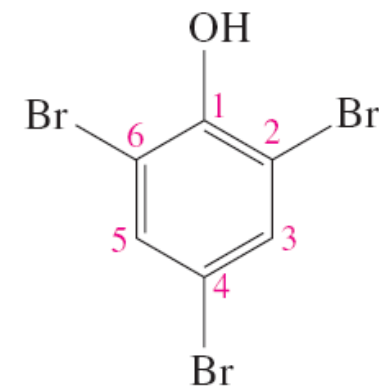
Positions indicated by numbers (1,2-, 1,3-, 1,4-) or prefixes: ortho- (2-), meta- (3-), para- (4-).

Polysubstituted Benzenes

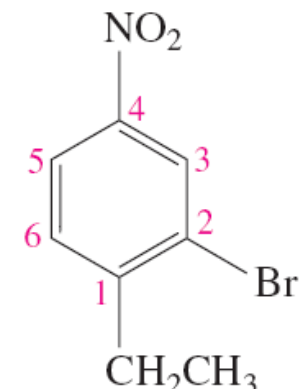
Positions numbered to give substituents the lowest possible numbers. When common names are used, substituent positions are indicated by numbers.



4-chloro-2-nitrotoluene



2,4,6-tribromophenol



2-bromo-1-ethyl-4-nitrobenzene

Correct nomenclature is essential for unambiguous identification of substituted aromatic compounds, especially in research publications and industrial applications.

Electrophilic Aromatic Substitution (EAS)



Electrophile Generation

Formation of an electrophilic species



π -Complex Formation

Electrophile interacts with π electrons



σ -Complex Formation

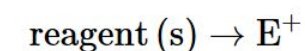
Formation of resonance-stabilized carbocation



Proton Loss

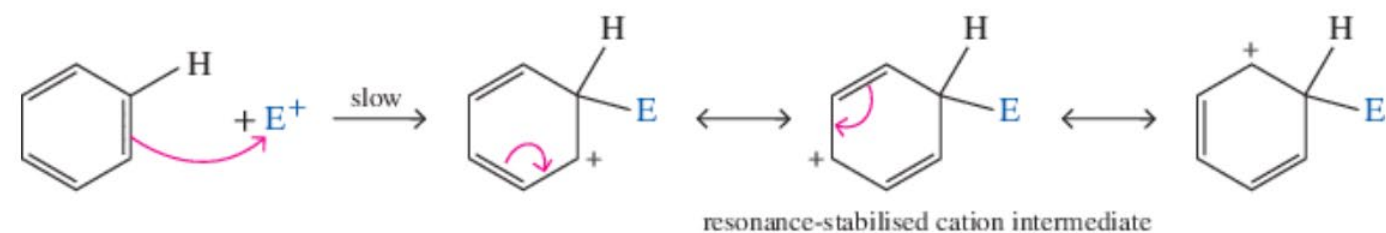
Deprotonation restores aromaticity

Step 1: Formation of the electrophile



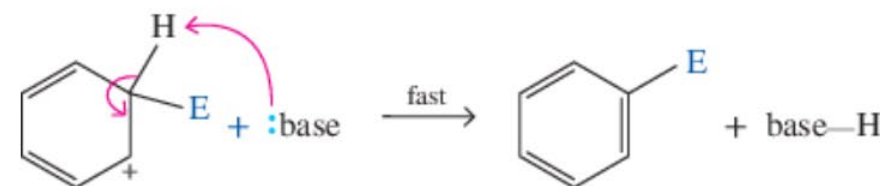
Step 2:

Reaction of the electrophile with the aromatic ring to give a resonance-stabilised cation intermediate



Step 3:

Proton transfer to a base to regenerate the aromatic ring

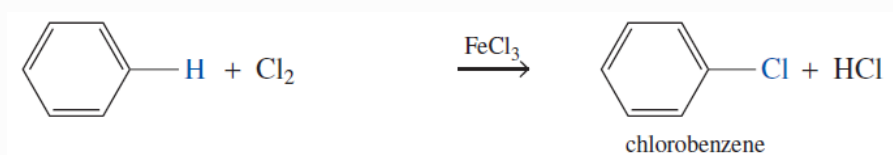


Electrophilic aromatic substitution proceeds through a resonance-stabilized carbocation intermediate (σ -complex). This pathway explains why aromatic compounds undergo substitution rather than addition reactions.

Common Electrophilic Aromatic Substitution Reactions

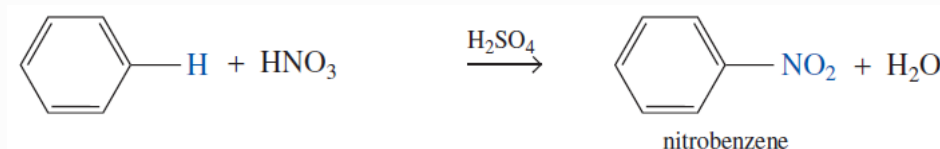
- **Halogenation**

Cl_2 or Br_2 with **Lewis acid catalyst** (AlCl_3 , FeBr_3) forms chlorobenzene or bromobenzene.



- **Nitration**

HNO_3 in H_2SO_4 generates the nitronium ion (NO_2^+) electrophile to form nitrobenzene.



- **Sulfonation**

Concentrated H_2SO_4 generates SO_3 or HSO_3^+ electrophiles to form benzenesulfonic acid.



Common Electrophilic Aromatic Substitution Reactions

Halogenation

Cl₂ or Br₂ with **Lewis acid catalyst**

Forms chlorobenzene or bromobenzene

Similar to thyroid hormone synthesis

Nitration

HNO₃ in H₂SO₄ generates NO₂⁺ electrophile

Forms nitrobenzene

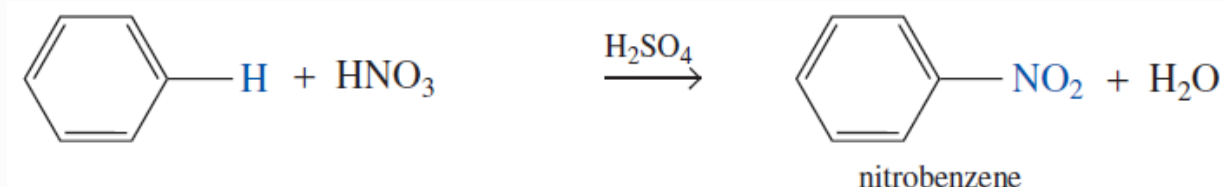
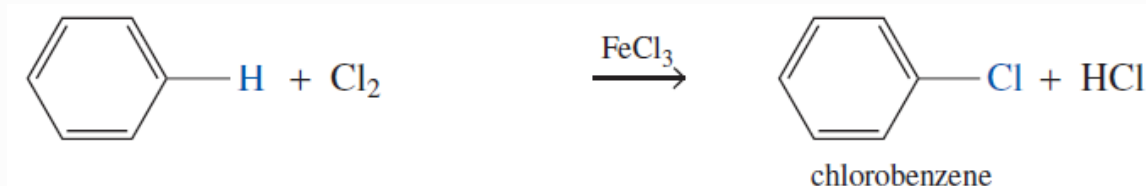
Analogous to nitration of tyrosine residues

Sulfonation

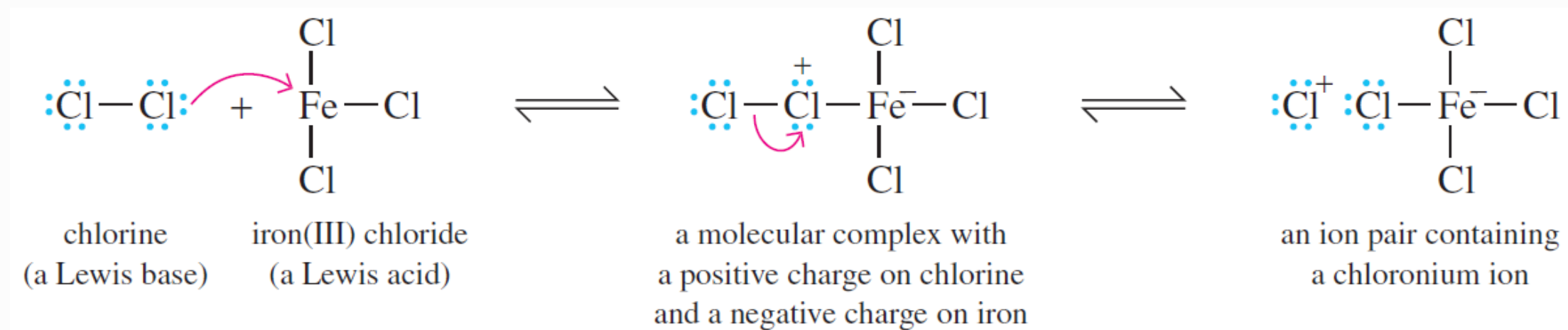
Concentrated H₂SO₄ generates SO₃ electrophile

Forms benzenesulfonic acid

Related to biological sulfation processes



EAS Mechanism: Step1 - Electrophile Formation



Halogenation

$X_2 + \text{Lewis acid} \rightarrow \text{polarized complex}$

AlCl_3 or FeBr_3 acts as catalyst

Creates partial positive charge on halogen

Nitration

$\text{HNO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{NO}_2^+ + \text{HSO}_4^- + \text{H}_2\text{O}$

Nitronium ion is potent electrophile

Similar to nitration in protein modification

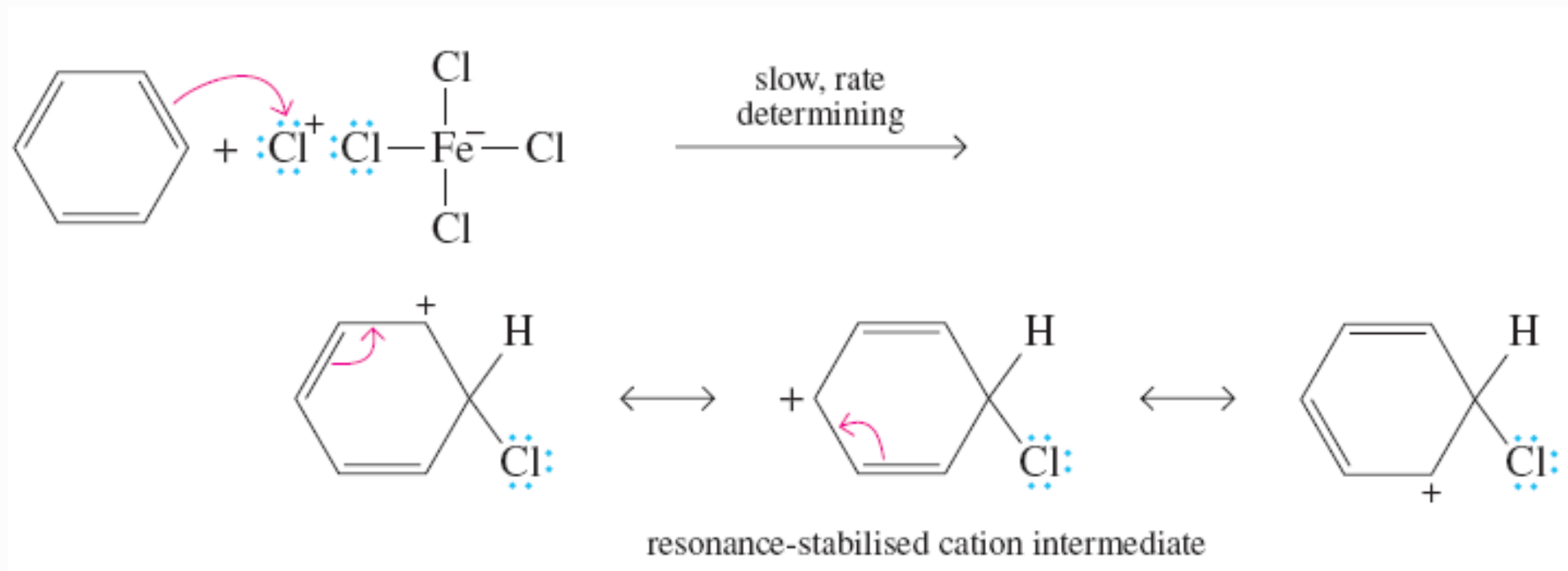
Sulfonation

$\text{H}_2\text{SO}_4 \xrightarrow{\text{heat}} \text{SO}_3 + \text{H}_2\text{O}$

SO_3 or HSO_3^+ acts as electrophile

Parallels biological sulfation pathways

EAS Mechanism: Step 2 Carbocation Formation



π -Complex Formation

Electrophile attracted to electron-rich π system

Forms weak initial complex

Reversible process

σ -Complex Formation

Electrophile forms covalent bond with carbon

Creates resonance-stabilized carbocation

Rate-determining step

Resonance Stabilization

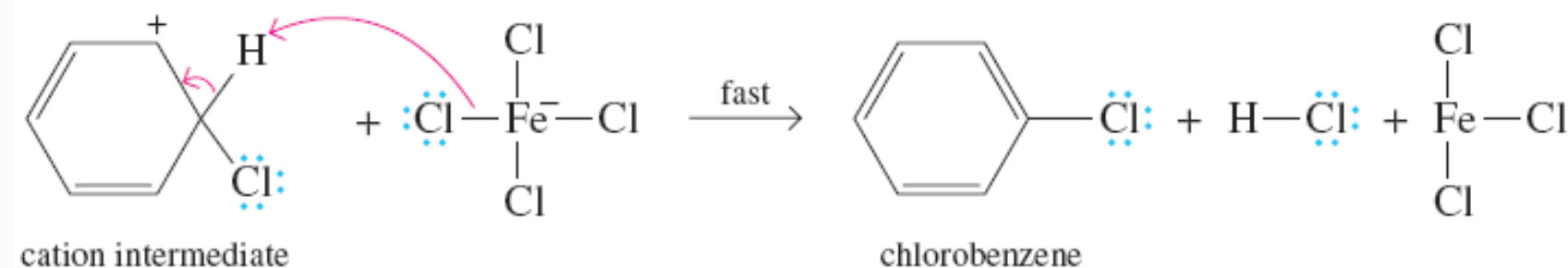
Positive charge distributes through ring

Aromatic character partially lost

Similar to enzyme transition state stabilization

EAS Mechanism:

Step 3 - Rearomatization



- 1** — Carbocation Intermediate
Resonance-stabilized but still high in energy due to loss of aromaticity
- 2** — Deprotonation
Base (typically X^- or solvent) removes proton from the carbon bearing the electrophile
- 3** — Rearomatization
The resulting electron pair restores the aromatic sextet
- 4** — Substituted Product
Aromatic ring with electrophile replacing one hydrogen atom

The driving force for this step is the restoration of aromaticity, which provides significant stabilization. The process is typically fast and irreversible, making electrophilic aromatic substitution a thermodynamically favorable process.

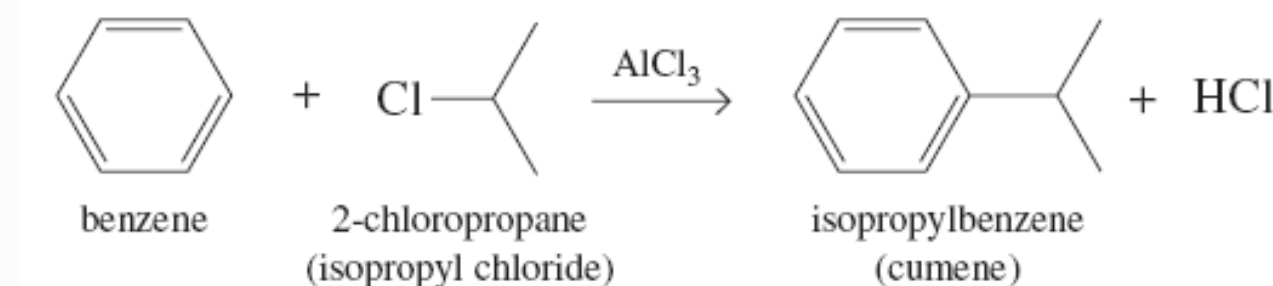
Friedel-Crafts Alkylation

Carbon-Carbon Bond Formation

Introduces alkyl groups onto aromatic rings, creating important C-C bonds.

Electrophile Generation

Alkyl halide reacts with AlCl_3 to form carbocation electrophile.



Limitations

Doesn't work with electron-withdrawing substituents on benzene.

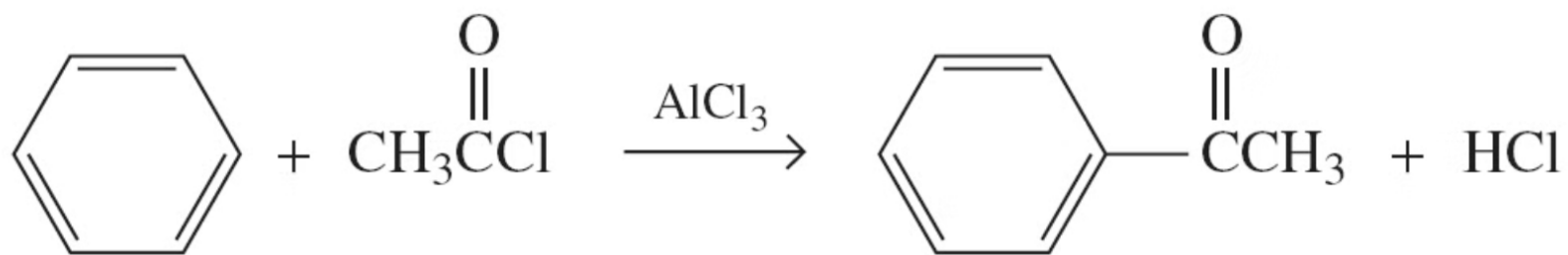


Rearrangements

Carbocations can rearrange, leading to product mixtures.

The Friedel-Crafts alkylation follows the general EAS mechanism and is a powerful method for introducing alkyl groups onto aromatic rings. However, polyalkylation can occur since alkyl groups are activating and ortho/para-directing.

Friedel-Crafts Acylation



Mechanism

Acyl halide + $\text{AlCl}_3 \rightarrow$
acylium ion ($\text{RC}=\text{O}^+$)



Advantages

No rearrangements,
only mono-acylation
occurs



Pharmaceutical Applications

Used in synthesis of
many aromatic drugs

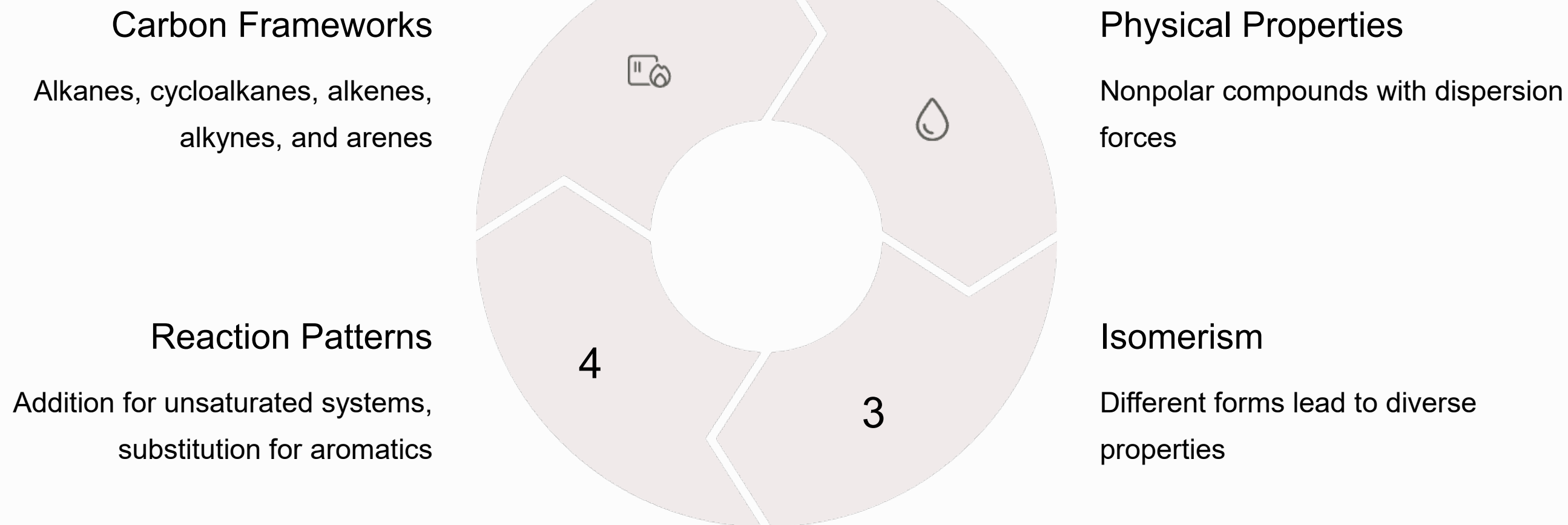


Versatility

Carbonyl group can
undergo further
transformations

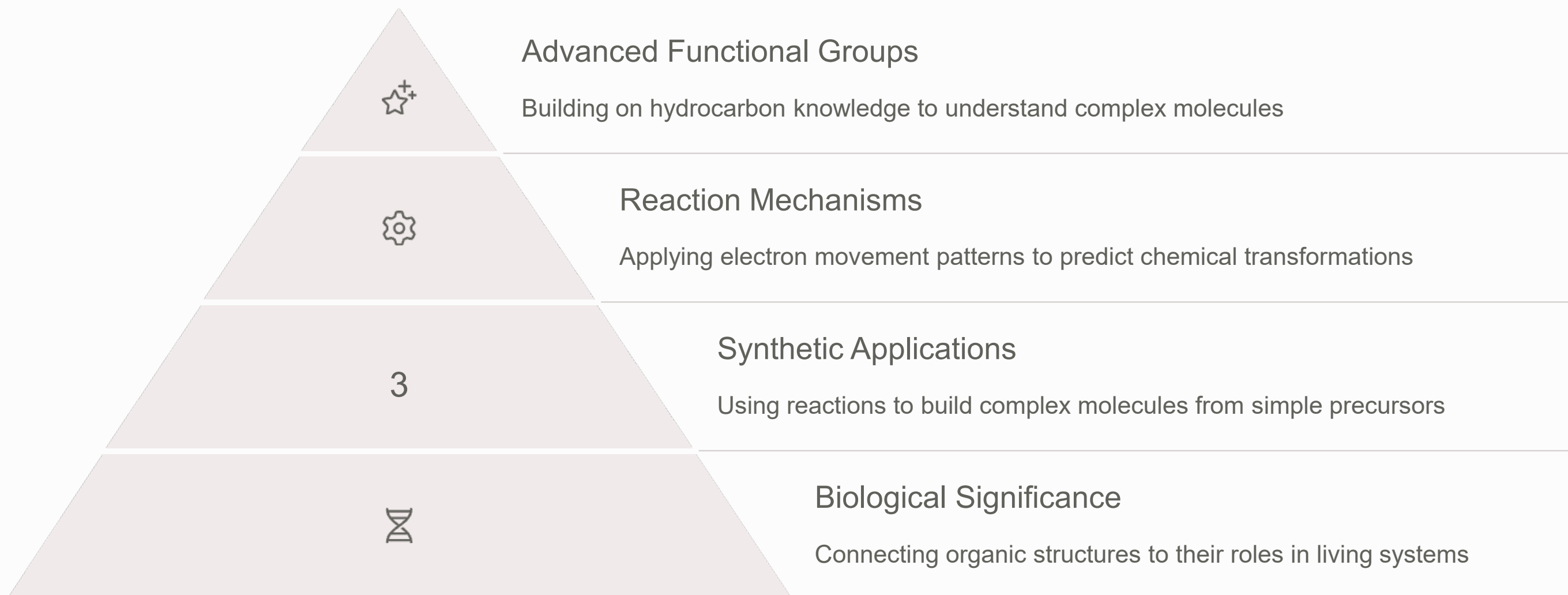
Friedel-Crafts acylation introduces a ketone functional group to the aromatic ring. The resulting aryl ketone is valuable in organic synthesis as it can undergo numerous subsequent transformations, including reduction to alkyl groups.

Summary: Chemistry of Carbon Frameworks



Understanding the fundamental structures and reactivity patterns of hydrocarbons provides the foundation for exploring more complex organic molecules with various functional groups.

Organic Chemistry: Looking Forward



The principles learned about hydrocarbons form the foundation for understanding more complex organic molecules with diverse functional groups and reaction patterns.

Common Functional Groups

	Family						
	Alkane	Alkene	Alkyne	Aromatic	Haloalkane	Alcohol	Ether
Functional group	C—H and C—C bonds			Aromatic ring			
	Ch 16	Ch 16	Ch 16	Ch 16	Ch 18	Ch 19	Ch 19
General formula	RH	RCH=CH ₂ RCH=CHR R ₂ C=CHR R ₂ C=CR ₂	RC≡CH RC≡CR	ArH	RX	ROH	ROR
Specific example	CH ₃ CH ₃	CH ₂ =CH ₂	HC≡CH		CH ₃ CH ₂ Cl	CH ₃ CH ₂ OH	CH ₃ OCH ₃
IUPAC name	Ethane	Ethene	Ethyne	Benzene	Chloroethane	Ethanol	Methoxymethane
Common name ^a	Ethane	Ethylene	Acetylene	Benzene	Ethyl chloride	Ethyl alcohol	Dimethyl ether

^aThese names are also accepted by the IUPAC.

	Family						
	Amine	Aldehyde	Ketone	Carboxylic Acid	Ester	Amide	Nitrile
	Ch 19	Ch 21	Ch 21	Ch 23	Ch 23	Ch 23	Ch 23
RNH ₂ R ₂ NH R ₃ N							
CH ₃ NH ₂							
Methanamine		Ethanal	Propanone	Ethanoic acid	Methyl ethanoate	Ethanamide	Ethanenitrile
Methylamine		Acetaldehyde	Acetone	Acetic acid	Methyl acetate	Acetamide	Acetonitrile

Functional Groups

	Family						
	Alkane	Alkene	Alkyne	Aromatic	Haloalkane	Alcohol	Ether
Functional group	C—H and C—C bonds			Aromatic ring			
	Ch 16	Ch 16	Ch 16	Ch 16	Ch 18	Ch 19	Ch 19
General formula	RH	RCH=CH ₂ RCH=CHR R ₂ C=CHR R ₂ C=CR ₂	RC≡CH RC≡CR	ArH	RX	ROH	ROR
Specific example	CH ₃ CH ₃	CH ₂ =CH ₂	HC≡CH		CH ₃ CH ₂ Cl	CH ₃ CH ₂ OH	CH ₃ OCH ₃
IUPAC name	Ethane	Ethene	Ethyne	Benzene	Chloroethane	Ethanol	Methoxymethane
Common name ^a	Ethane	Ethylene	Acetylene	Benzene	Ethyl chloride	Ethyl alcohol	Dimethyl ether

Funktionelle grupper er ekstremt vigtige i organisk kemi...

lær dem!

Common

RNH ₂ R ₂ NH R ₃ N						RCN
CH ₃ NH ₂	CH ₃ CHO	CH ₃ C(=O)CH ₃	CH ₃ COOH	CH ₃ COCH ₃	CH ₃ C(=O)NH ₂	CH ₃ C≡N
Methanamine	Ethanal	Propanone	Ethanoic acid	Methyl ethanoate	Ethanamide	Ethanenitrile
Methylamine	Acetaldehyde	Acetone	Acetic acid	Methyl acetate	Acetamide	Acetonitrile